

RESEARCH ARTICLE OPEN ACCESS

Assessing and Explaining Human–Nature Coordination at the County Scale in China: Combining an Interpretable Spatiotemporal Graph Neural Network and Difference-In-Differences

Qi Liu¹  | Yirong Fan¹ | Mengjuan Li¹ | Bo Li¹ | Jiajun Qiao²

¹Faculty of Geographical Science, Beijing Normal University, Beijing, China | ²College of Geographical Sciences, Henan University, Zhengzhou, China

Correspondence: Bo Li (libo@bnu.edu.cn)

Received: 6 January 2026 | **Revised:** 14 July 2026 | **Accepted:** 20 July 2026

Keywords: human–nature coordination | integrated gradients | MTGNN | multi-period DID | NKEFZ

ABSTRACT

Coordinating high-intensity human activities with fragile ecosystems remains a major challenge for global sustainability. Accurate assessment of human–nature coordination (HNC) is critical, yet county-level variations, underlying drivers, and responses to ecological zoning have seldom been examined simultaneously within a unified framework. We construct an HNC index for Chinese county-level units from 2001 to 2022 by integrating local and telecoupling coordination degree between ecosystem-service supply and human activity. For 1817 counties with complete covariates, a multivariate time series graph neural network (MTGNN), integrated gradients (IG), and partial dependence plots (PDPs) characterize temporal and cross-county predictive dependencies. A staggered difference-in-differences (DID) design then estimates the effect of National Key Ecological Function Zones (NKEFZ). National HNC declined slightly after 2016; 45.0% of counties improved and 54.7% deteriorated over the full study period. Precipitation, temperature, and slope contributed 34.7%, 12.3%, and 10.3% of model attribution, respectively, and precipitation was the dominant predictor in 66.1% of counties. The learned graph further identified nonlocal predictive communities. NKEFZ increased standardized HNC by 0.034. Formal channel tests support ecological restoration, reduced industrial activity pressure, and public-service support, with the NDVI pathway accounting for approximately 30.9% of the total effect; the industrial-structure pathway was not statistically supported. Policy effects differed across regions and limiting-factor groups, while heterogeneity among learned-graph communities remained relatively low. These findings connect county-level HNC dynamics, predictive constraints, and ecological-zoning effects, providing evidence for constraint-sensitive policy design.

Abbreviations: ADV, industrial structure advancement; AI, aggregation index; AVSI, added value of the secondary industry; CLP, cultivated land proportion; CSDID, Callaway and Sant' Anna difference-in-differences; DEM, elevation; DID, difference-in-differences; EC, east coast; ESI, ecosystem service supply index; HAI, human activity index; HNC, human–nature coordination index; IG, integrated gradients; ISP, impervious surface proportion; LFE, local fiscal general budget expenditure; LTCCD, local and telecoupling coordination degree; MTGNN, multivariate time series graph neural network; MYG, Middle Yangtze River; MYW, Middle Yellow River; NC, North Coast; NDVI, normalized difference vegetation index; NE, Northeast; NKEFZ, national key ecological function zones; NW, Northwest; PDP, partial dependence plot; PFE, public service support proxy; PRE, annual cumulative precipitation; PRS, per capita retail sales; REP, rural employed persons; SC, South Coast; SHAPE, mean shape index; SHDI, Shannon's diversity index; SHEI, Shannon's Evenness index; SLOPE, slope; SW, Southwest; TFA, total social fixed asset investment; TMP, annual mean temperature.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](https://creativecommons.org/licenses/by-nc/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

© 2026 The Author(s). *Land Degradation & Development* published by John Wiley & Sons Ltd.

1 | Introduction

Global warming and rapid urbanization are profoundly reshaping human–nature interactions, while the tension between ecosystem degradation and growing human development demands has emerged as a major constraint on sustainable development (Wei et al. 2024). As the largest developing country, China is currently at a critical stage characterized by the simultaneous pursuit of ecological conservation and economic growth (Zhang and Wang 2024). From a climatic perspective, China is characterized by a predominantly warm and humid climate, with frequent flood-related hazards, and many regions have experienced record-breaking or highly destructive extreme events, such as heatwaves and intense rainfall. From a socio-economic perspective, China has undergone the largest and fastest urbanization process worldwide, during which urban land area expanded by nearly 2.7 times (Ministry of Housing and Urban–Rural Development of China 2025) over the past two decades (by the end of 2023), and the national urbanization rate reached 67% in 2024 (Xinhua News Agency 2025). Against this background, China encounters the global challenge of reconciling ecological conservation with increasing human development demands, while its vast territory, diverse ecosystems, and substantial regional disparities generate highly heterogeneous spatial patterns and differentiated trajectories of human–nature interactions.

Accurately quantifying human–nature relationships is essential for advancing the understanding of human–nature system dynamics. Existing studies have investigated human–nature relationships from multiple perspectives, including climate change (Chen et al. 2023; Li et al. 2024; Zeng et al. 2022), urban expansion (Liu et al. 2025; Qi et al. 2023), ecological engineering (Fan and Wei 2025), and land-use and resource interactions. As research has progressed, increasing attention has been paid to the human–nature relationship from a bidirectional interaction perspective, including land-use efficiency (Su et al. 2024), the coupling between grazing activities and ecological conditions (Liao et al. 2023), as well as integrated nexus frameworks such as the water–energy–food system (Karri et al. 2026; Yang et al. 2024) and carbon–water coupling (Li, Chen, et al. 2025). However, most existing coordination assessments remain focused on interactions between two subsystems within individual regions, with limited consideration of potential non-local dependencies among counties. Distance, shared environmental change, and resource linkages can connect nonadjacent counties. We therefore use a local and distance-weighted remote component to represent potential cross-county coordination.

Human–nature interactions constitute a complex and dynamic process, the essence of which lies in elucidating the mechanisms through which human activities interact with ecosystems (Fu 2020; Lu 2003). Existing studies have largely focused on directly quantifiable indicators and employed a range of analytical approaches—such as panel data regression models (Hu et al. 2024), geographic detectors (Fan et al. 2022), geographically weighted regression (Kong et al. 2026), and machine learning models (Liu, Qiao, Li, and Huang 2023)—to investigate the driving forces of human–nature coordination (HNC). Among these approaches, machine learning models exhibit notable advantages in capturing nonlinear relationships and have emerged as a prominent frontier in this field. However, conventional

machine learning methods are limited in their ability to simultaneously capture temporal dynamics and spatial dependencies. In contrast, spatiotemporal graph neural networks, by explicitly coupling spatial associations with temporal evolution, offer a novel means to disentangle complex interaction processes. Nevertheless, such models remain limited in capturing policy effects, resulting in a persistent gap between the attribution of driving forces and the evaluation of policy impacts. Therefore, integrating spatiotemporal graph neural networks with policy effect evaluation represents a promising methodological direction for addressing this challenge.

The National Key Ecological Function Zones (NKEFZ) policy represents a nationwide ecological zoning initiative, and a growing body of research has documented its multidimensional effects. From an environmental perspective, the NKEFZ policy has been shown to improve regional air quality (Li et al. 2021), enhance water quality and vegetation cover (Fan et al. 2024), and strengthen overall ecological functions (Xu et al. 2025). In terms of social development, the policy has effectively enhanced the provision of basic public services at the county level (Chen et al. 2024) and contributed to narrowing the urban–rural income gap (Lu and Zhang 2023). With respect to economic outcomes, existing studies indicate that NKEFZ has increased rural household income (Wang et al. 2024), fostered the development of green industries (Chang et al. 2025), and supported higher-quality economic growth (Zhang et al. 2025). However, studies that conceptualize the human–nature relationship as an integrated system and examine the impacts and underlying pathways of the NKEFZ policy remain limited. Moreover, although counties constitute a critical scale for both policy implementation and human–nature interactions, analyses of HNC at the county level are scarce. This limitation constrains a systematic understanding of county-level HNC and hampers the identification of the specific mechanisms through which NKEFZ influences HNC.

Against the above research gaps, this paper carries out four strands of analysis as follows: (1) Reveal the spatiotemporal dynamics and evolutionary patterns of human–nature coordination (HNC) at the county scale from 2001 to 2022; (2) Identify the dominant driving factors, nonlinear response patterns, and spatial heterogeneity of HNC via interpretable spatiotemporal graph neural networks; (3) Quantitatively evaluate the policy effects and governance pathways of national key ecological function zones (NKEFZ); (4) Synthesize all analytical findings to put forward differentiated policy suggestions for human–environment sustainable development tailored to various geographic regions.

The main innovations of this study lie in two dimensions. First, it extracts the core evolutionary patterns of county-level HNC across China using time-series clustering. Second, this paper constructs an integrated MTGNN-IG-DID modeling framework. This framework can simultaneously capture the time-lag effects and spatial heterogeneity of HNC driving forces, as well as quantify the policy impacts generated by the NKEFZ initiative on HNC. By coupling mechanism decomposition and policy assessment, the established framework deepens theoretical cognition regarding human–nature coordination mechanisms and delivers replicable theoretical and methodological references for regionally targeted governance and sustainable development policy formulation.

2 | Materials and Methods

In recent decades, China has undergone an exceptionally rapid urbanization process, unprecedented in scale, during which the disorderly expansion of built-up land, abrupt shifts in land-use patterns, and increasing ecosystem fragmentation have emerged as major constraints on sustainable and high-quality development. Given the pronounced geographical and socioeconomic disparities across regions in China, the country is divided into eight regions in this study (Figure 1a) to facilitate subsequent analyses of spatial heterogeneity. Detailed regional classifications are provided in Appendix S1.

As a key spatial policy instrument for balancing ecological conservation and development, ecological zoning has been indispensable in strengthening the national ecological security barrier through a range of measures, including ecological transfer payments, industrial access restrictions, and ecological migration and resettlement programs. China implemented two rounds of zoning adjustments and designations for NKEFZs in 2010 and 2016 (Figure 1b),

ultimately establishing zones that cover approximately 53% of the country's land area. Nevertheless, the spatiotemporal dynamics of human–nature interactions at the county scale remain insufficiently characterized, limiting deeper insights into human–nature relationships and complicating the development of region-specific sustainable development strategies.

2.1 | Research Framework

The analytical framework of this study consists of five sequential steps (Figure 2): first, we integrate remote sensing products, statistical yearbooks, and policy archives to construct county-level panel datasets covering ESI, HAI, driving factors, and the delineation of NKEFZ; second, we adopt the Local and telecoupling coordination degree (LTCCD) model to calculate ESI and HAI values and further derive the HNC index based on the LTCCD outputs; third, we employ trend analysis, spatial autocorrelation, and Mfuzz clustering to characterize the spatial patterns and temporal trajectories of HNC; fourth, we implement spatiotemporal prediction via

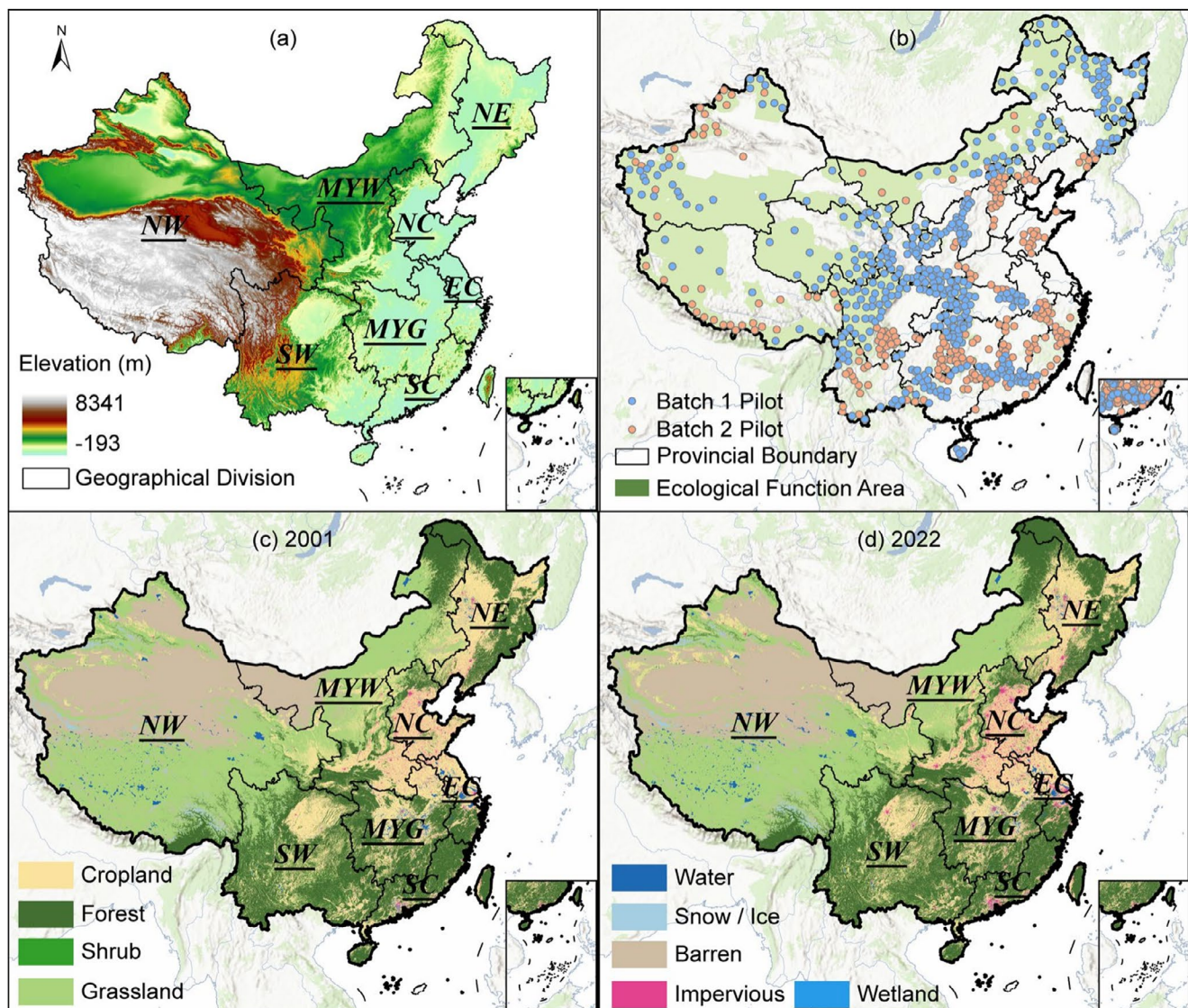


FIGURE 1 | Study area. Wiley acknowledges that the borders within the figure are subject to multiple territorial claims. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.70834)]

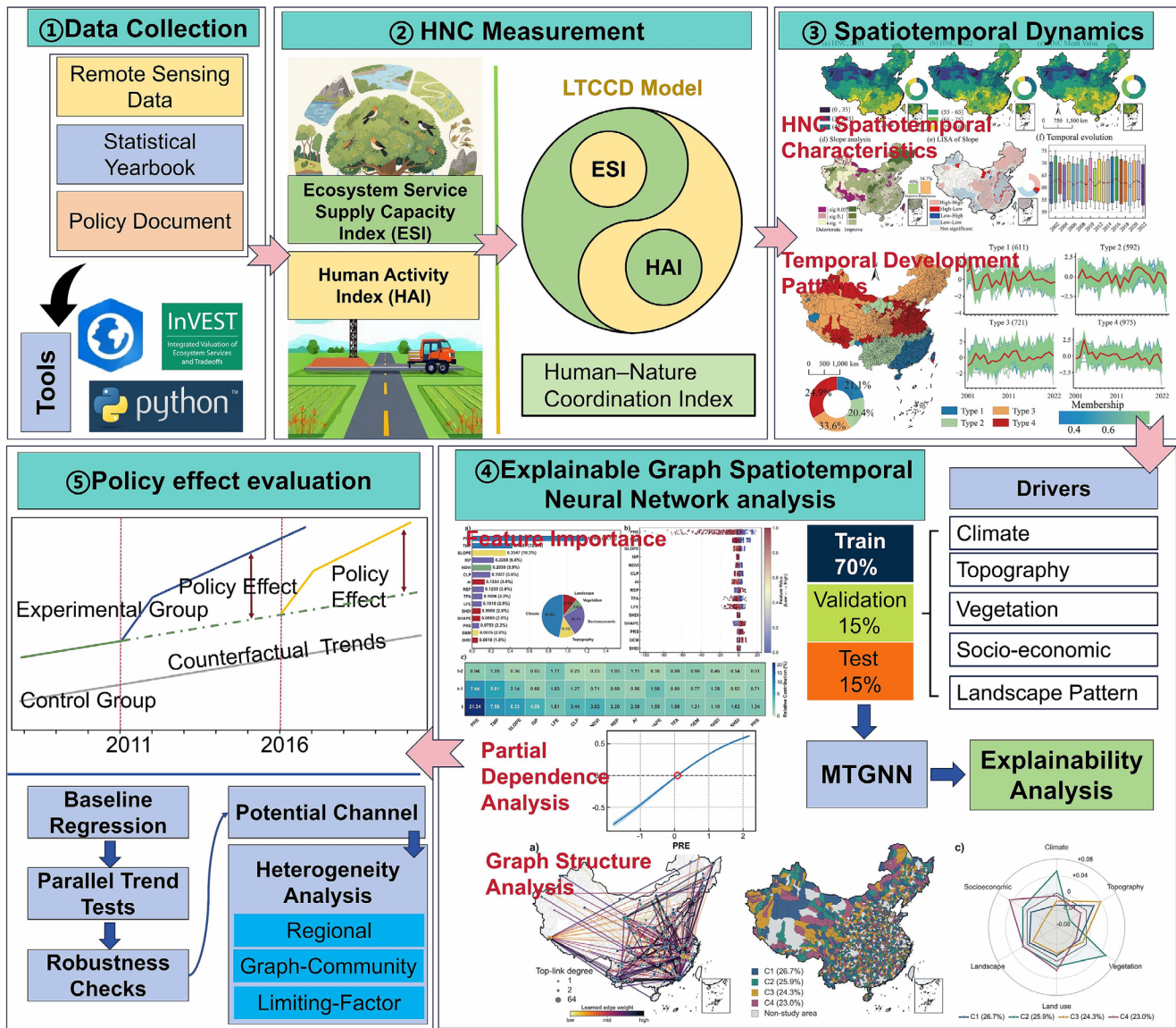


FIGURE 2 | Research framework. Wiley acknowledges that the borders within the figure are subject to multiple territorial claims. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.70834)]

the MTGNN model and interpret variable contributions, lagged response relationships, and inter-county correlation structures through integrated gradients (IG), partial dependence plots (PDP), and adaptive graph learning; fifth, we treat the phased rollout of NKEFZ as a quasi-natural experiment and evaluate its policy effects using baseline DID, event study design, CSDID, robustness checks, mechanism channel tests, and multi-layer heterogeneity analysis. Ultimately, this study synthesizes HNC evolutionary characteristics, predictive driving mechanisms, and policy intervention effects to deliver empirical evidence supporting differentiated county-scale governance.

2.2 | Measurement of ESI and HAI

The ESI is calculated by integrating four key ecosystem services in China: water conservation, soil retention, biodiversity, and windbreak and sand fixation, and it reflects the ecosystem's ability to provide services. Each ecosystem service is normalized

to 0–100, then weighted analysis is performed at the grid scale (1 km), and finally the average value is obtained through county-level zonal statistics. Its calculation formula is as follows:

$$ESI = \frac{1}{n} \sum_{i=1}^n w_i ES_i \quad (1)$$

where ES denotes an ecosystem service, i signifies one of the four key services for a given region, and w is the corresponding weight. Following established methodology (Wang et al. 2025), all four services were deemed equally important, and each was assigned a weight of 25%. Detailed computational procedures of ecosystem services are provided in Appendix S2.

The HAI is derived from the dataset developed by Mu et al. (2022), which integrates eight variables—built environment, population density, nighttime lights, cropland, pasture, roads, railways, and navigable waterways—to effectively characterize the dynamics of the human footprint. This dataset has

been widely applied in related studies (Li, Du, et al. 2025; Liu, Qiao, Li, and Huang 2023; Xia et al. 2023).

2.3 | Measurement of HNC

The measurement of HNC covers three components: local and remote coupling degrees, local and remote comprehensive development levels, and the final coordination index. HNC quantifies the coordination status between these two elements in both local contexts and potential long-distance spatial linkages.

The local component captures the coupling relationship between ESI and HAI within a single county, whereas the remote component constructs decay weights based on the centroid distances of counties to reflect potential spatial connections between a target county and other regions. The detailed calculation procedures are presented as follows: (Zhang et al. 2023).

First, the local and remote coupling degree C_{si} is calculated in accordance with (An et al. 2023):

$$C = \sqrt{\frac{E \times H}{(E + H)/2}} \quad (2)$$

$$C_{si} = \mu \cdot C(E_i, H_i) + \lambda \left(\frac{\sum_{k=1}^n w_{ik} \times C(E_i, H_k) + \sum_{k=1}^n w_{ki} \times C(E_k, H_i)}{2} \right) \quad (3)$$

$$w_{ik} = \frac{d_{ik}^{-p}}{\sum_{k=1}^n d_{ik}^{-p}} \quad (4)$$

In Equation (2), E and H stand for ESI and HAI respectively, and C denotes the local coupling degree. In Equation (3), μ and λ refer to the weights assigned to the local and remote coupling components, both set to a baseline value of 0.5; i represents the target county, and n is the total number of counties. In Equation (4), w_{ik} is the spatial weight constructed via the inverse distance method, d_{ik} denotes the centroid distance between county i and county k , and the distance decay parameter p is fixed at 2.

Second, the local and remote comprehensive development level T_{si} is calculated:

$$T = W_1 \times E + W_2 \times H \quad (5)$$

$$T_{si} = \mu T(U, E) + \lambda \left(\frac{\sum_{k=1}^n w_{ik} \times T(E_i, H_k) + \sum_{k=1}^n w_{ki} \times T(E_k, H_i)}{2} \right) \quad (6)$$

In Equation (5), T represents the comprehensive development level of ESI and HAI. The baseline setup assumes equal importance of the two subsystems, so both W_1 and W_2 are assigned a

value of 0.5. The weight settings for local and remote components in Equation (6) are consistent with those in Equation (3).

Third, integrate C_{si} and T_{si} to derive the HNC index:

$$HNC_{si} = \sqrt{C_{si} \times T_{si}} \times 100 \quad (7)$$

We conducted a sensitivity analysis of HNC using different weighting schemes for the local and remote components (Appendix S6). The results reveal highly consistent patterns across all weighting configurations, indicating that the allocation of weights does not materially affect the main findings. Finally, HNC values are classified according to their data characteristics as follows: (Table 1).

2.4 | Time-Series Cluster Analysis

The R package *Mfuzz*, which is based on the fuzzy C-means (FCM) clustering algorithm (Lokesh and Matthias 2019), is employed to identify the temporal evolutionary patterns of HNC. This method has been extensively validated in fields such as bioinformatics and gene expression analysis (Shen et al. 2020). The FCM algorithm represents a fuzzy extension of the K-means clustering approach, in which classification is performed using membership functions rather than hard assignments. By determining the evolutionary pattern of each county according to its membership degree, FCM enhances the capability to handle fuzziness and uncertainty compared with the traditional K-means algorithm. The core procedure of FCM clustering involves the iterative optimization of an objective function to obtain the final clustering results. The objective function is formulated as follows (Liu, Qiao, Han, et al. 2023; Liu et al. 2024):

$$J(MC) = \sum_{i=1}^n \sum_{j=1}^c \mu_{ij}^\varphi d_{ij}^2 \quad i = 1, \dots, n; \quad j = 1, \dots, c \quad (8)$$

In the equation, C denotes the centroid matrix of the clusters, and M represents the fuzzy membership matrix of HNC with a dimension of $n \times c$. M_{ij} ($\mu_{ij} \in [0, 1]$) indicates the degree of membership between the i county and the j cluster, while $\varphi (\varphi \geq 1)$

TABLE 1 | Classification criteria for HNC.

Category	HNC	Coordination category
Uncoordinated development	(0, 35]	Severe coordination recession category
Transitional development	(35, 45]	Moderate coordination recession category
	(45, 55]	Near-disorder coordination category
Coordinated development	(55, 65]	Primary coordination development category
	(65, 75]	Moderate coordination development category
	(75, 100]	High coordination development category

denotes the fuzzification exponent. The term d_{ij}^2 represents the squared distance between the i county and the j cluster.

2.5 | Multivariate Time Series Graph Neural Network (MTGNN)

The MTGNN is a deep learning model specifically designed for multivariate time series forecasting tasks with complex spatio-temporal dependencies. Originally proposed by (Wu et al. 2020), its key innovation lies in the ability to adaptively learn spatial dependency relationships among variables without requiring a predefined graph structure or spatial weight matrix. Traditional spatiotemporal prediction methods typically rely on predefined adjacency matrices (e.g., based on geographic distance or administrative boundaries), which often fail to capture complex and implicit inter-variable relationships. By adopting an end-to-end learning framework, MTGNN simultaneously models temporal dynamics and spatial interactions, effectively overcoming this limitation.

The MTGNN architecture consists of three core modules. First, the Graph Learning Layer learns a pair of embedding vectors (source and target embeddings) for each node and adaptively constructs the adjacency matrix by computing the similarity between these embeddings. The resulting asymmetric graph structure enables the model to capture directed spatial influence relationships. Second, the Graph Convolution Module performs information aggregation based on the learned graph structure. By employing a mix-hop propagation mechanism, the model aggregates neighborhood information across multiple propagation depths, thereby extracting multi-scale spatial features. Third, the Temporal Convolution Module adopts dilated causal convolution to capture multi-scale temporal dependencies. Through exponentially increasing dilation rates, the model achieves a long temporal receptive field with a relatively small number of layers while preserving causality, ensuring that predictions depend solely on historical information. The model architecture and experimental codes are available at <https://github.com/Albert-007MAX/MTGNN-panel-regression-code>.

MTGNN is particularly suitable for the present HNC analysis because county-level HNC reflects both temporally lagged responses to multiple climatic and socioeconomic drivers and latent cross-county associations that cannot be fully represented by a fixed geographic adjacency matrix. The graph-learning layer infers data-driven inter-county dependencies, while mix-hop propagation aggregates information from direct and higher-order learned neighbors; together with temporal convolutions, these components capture the multiscale spatiotemporal structure relevant to county-level HNC prediction.

In this study, MTGNN is applied to analyze the multidimensional driving factors of HNC. In terms of data organization, 1817 counties within the study area are treated as graph nodes to construct a spatiotemporal panel dataset. At each time step, each node is characterized by 15 driving factors as input features, covering five major categories: climate, vegetation characteristics, topography, landscape pattern factors, and socioeconomic factors (Table 2). For reasons of data continuity and availability, this study adopts 1817 county-level units, as detailed

TABLE 2 | Selection of driving factors.

Dimension	Selected indicators	Abbreviation
Climate	Annual cumulative precipitation	PRE
	Annual mean temperature	TMP
Topography	Elevation	DEM
	Slope	SLOPE
Vegetation	Vegetation coverage	NDVI
Socioeconomic	Cultivated land proportion	CLP
	Impervious surface proportion	ISP
	Local fiscal general budget expenditure (10,000 yuan)	LFE
	Total social fixed asset investment (10,000 yuan)	TFA
	Per capital total retail sales of consumer goods	PRS
	Rural employed persons (Person)	REP
Landscape pattern	Mean shape index	SHAPE
	Shannon's diversity index	SHDI
	Shannon's evenness index	SHEI
	Aggregation index	AI

in Appendix S3. Descriptions of the selected indicators and their data sources are provided in Appendix S4.

All input features are standardized using Z-score normalization prior to model training to eliminate the influence of scale differences on model learning. The detailed model configuration is illustrated in Figure 3 and described in Appendix S5.

Given the “black-box” nature of deep learning models, which limits their interpretability in scientific research, this study introduces Integrated Gradients (IG) (Sundararajan et al. 2017) to interpret the prediction results of MTGNN. IG is a gradient-based attribution method that quantifies the marginal contribution of each feature to the model output by computing the path integral of gradients from a baseline input (a zero vector) to the actual input. Compared with conventional gradient-based methods, IG satisfies the completeness axiom, whereby the sum of feature attributions equals the difference between the model output and the baseline output, thereby ensuring the reliability of the attribution results. In addition, Partial Dependence Plots

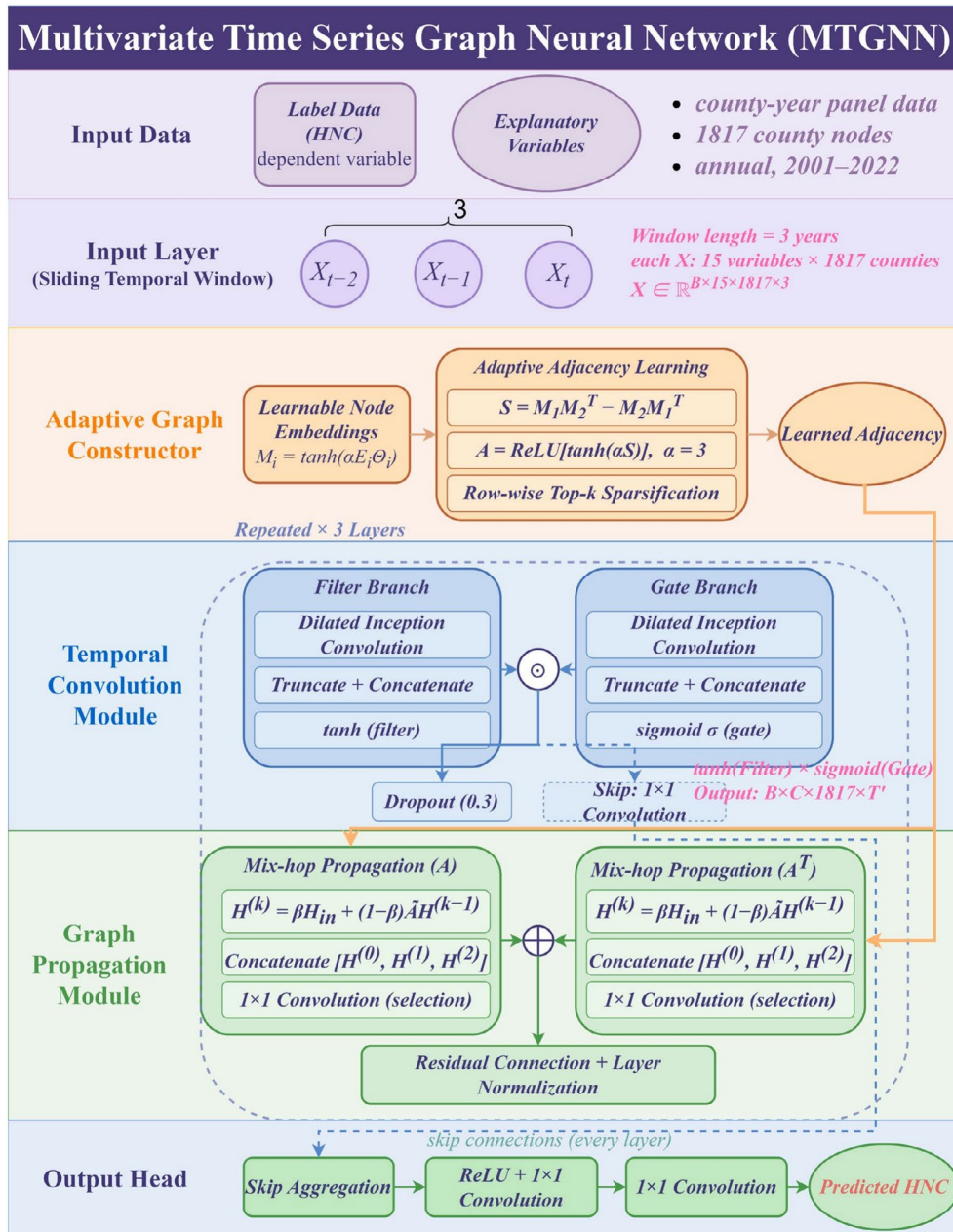


FIGURE 3 | MTGNN model architecture. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.20834)]

(PDPs) are employed to examine the nonlinear response relationships between individual driving factors and HNC. PDPs characterize the average effect of a target feature on model predictions by varying its values while taking the marginal expectation over the remaining features.

To further investigate the spatial organization embedded in the learned graph structure, we extracted the adaptive relational structure from the trained model and spatially visualized the high-weight inter-county connections. The Louvain community detection algorithm was then applied to the learned graph to identify clusters of counties. Subsequently, the combinations of explanatory variables were compared across clusters to characterize the distinct types of inter-county associations learned by the model during prediction and to reveal their corresponding attribute differences.

2.6 | Difference-In-Differences (DID) Model

DID approach is a cornerstone quasi-experimental method in policy evaluation. It aims to identify treatment effects by comparing the outcome changes over time between a treatment group and a control group, thereby helping to mitigate endogeneity concerns such as selection bias (Callaway and Sant'Anna 2021; Yin and Guo 2021). The variables in the DID framework are specified as follows: The treatment group indicator (treat) distinguishes between policy intervention and control groups, where county-level units within NKEFZ boundaries (treatment group) are assigned a value of 1 and counties outside NKEFZ (control group) receive 0. The policy timing variable (time) marks the implementation period, with the first policy cohort starting in 2011 (following the official announcement in December 2010) and

the second cohort in 2016, where years from the policy shock onward are coded as 1 and preceding years as 0. The core explanatory variable (NKEFZ) represents the interaction term between the treatment and time variables ($NKEFZ = \text{treat} \times \text{time}$), whose coefficient captures the net policy effect of NKEFZ on HNC and constitutes the key parameter of interest. The regression model is specified as follows (Fan and Yang 2024):

$$HNC_{it} = \beta_0 + \beta_1 NKEFZ_{it} + \gamma X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (9)$$

where i and t denote county unit and year, respectively; X_{it} represents a vector of control variables for county i in year t (selected based on preceding driving factor analysis); λ_t represents year fixed effects; μ_i represents county-specific fixed effects; and ε_{it} is the idiosyncratic error term. The coefficient β_1 captures the net policy effect of the core explanatory variable $NKEFZ_{it}$. β_0 is the constant term, and γ is the coefficient vector of control variables. To address potential post-treatment bias and align with the feature importance results from MTGNN, the control variables primarily consist of exogenous climatic factors, namely PRE and TMP.

The NKEFZ policy incorporates multiple instruments, including spatial land-use regulation, industrial access restrictions, ecological conservation and restoration, and ecological transfer payments. Its overarching objective is to maintain critical ecosystem functions and promote coordination between county-level development activities and regional resource–environmental carrying capacity. Based on this objective, the policy may influence HNC through three major pathways: ecological restoration, industrial restructuring, and fiscal public service compensation. Accordingly, this study examines the underlying mechanisms from these three perspectives.

Ecological restoration represents a direct pathway through which the policy affects HNC (Hao et al. 2019). NKEFZs implement a series of measures, including natural forest conservation, soil and water conservation, degraded ecosystem restoration, and ecological environmental quality assessments, to enhance ecosystem stability, improve ecological carrying capacity, and alleviate anthropogenic pressures on ecosystems. In this study, the normalized difference vegetation index (NDVI) is used as a proxy indicator for the ecological restoration pathway.

Industrial restructuring reflects the policy-induced constraints on regional development intensity (Gong et al. 2021). By establishing industrial access thresholds, regulating construction land expansion, and restricting highly polluting industries, NKEFZ policies aim to reduce resource consumption, land development pressure, and pollutant emissions associated with intensive industrial activities (Cheng et al. 2023; Zhang et al. 2024). This study uses secondary industry value added (AVSI) to represent the scale of industrial activities and the ratio of tertiary industry value added to secondary industry value added (ADV) to characterize the advancement of the industrial structure. A decline in AVSI after policy implementation would indicate that intensive industrial activities have been effectively constrained, whereas an increase in ADV would suggest a transition toward economic sectors associated with lower ecological pressures.

Fiscal public service compensation captures the redistribution effect of policy implementation on the costs of local ecological protection. Counties responsible for ecological conservation often face constraints arising from restricted development opportunities and limited fiscal capacity (Cao et al. 2021; Wang et al. 2024). Ecological transfer payments and related fiscal arrangements can compensate for the opportunity costs of conservation efforts while supporting essential public services, including education, healthcare, and infrastructure maintenance. In this study, public fiscal expenditure (PFE) is employed as a proxy for local fiscal capacity in providing public services. An increase in PFE following policy implementation would indicate that fiscal compensation has strengthened local governments' capacity to sustain public services and socioeconomic development under ecological protection constraints, thereby contributing to the stabilization of residents' welfare foundations.

2.7 | Data Sources

The data used in this study include the cartographic data of the study area, vector data of China's administrative divisions, ESI and HAI measurement data, driver data, and their related socioeconomic statistical data. Table 3 lists the sources of each dataset. The map verification number for the vector maps of China's provincial- and county-level administrative divisions is GS (2019) 1822, and the base map data remain unmodified. All data are resampled to a 1 km × 1 km resolution to match the land-use data and standardized to the Krasovsky 1940 Albers projection coordinate system.

3 | Results

3.1 | Spatiotemporal Dynamics of ESI and HAI

Temporally, the ESI in the study area exhibits an overall upward trend with fluctuations, with an average annual growth rate of 0.113 ($p < 0.05$). Nevertheless, this trend is characterized by clear stage-specific differences: the ESI declined during 2001–2011 and recovered from 2011 to 2016. From 2016 to 2022, the ESI displayed a fluctuating pattern, with an initial decrease followed by a subsequent increase (Figure 4g). Spatially, the ESI is lower in northern than in southern regions and lower in western than in eastern regions, forming an overall gradient that increases from the northwest to the southeast (Figure 4a–c).

Over the same period, the HAI exhibits a clear linear increasing trend, with an average annual growth rate of 0.286 ($p < 0.01$) (Figure 4h). Spatially, the HAI displays a pronounced gradient increasing from the northwest to the southeast, with markedly higher values observed in eastern coastal areas and provincial capital cities (Figure 4d–f).

3.2 | Spatiotemporal Dynamics of HNC

During the study period, the national HNC generally remained at the level of primary coordination development (Figure 5f), with

TABLE 3 | Data sources.

Data	Spatial resolution	Data source
Land use/Land cover (LULC)	30 m	http://doi.org/10.5281/zenodo.4417809
Annual Cumulative Precipitation (PRE)	1000 m	https://www.geodata.cn/data/datadetails.html?dataguid=192,891,852,410,344&docId=491
Potential evapotranspiration(Pet)	1000 m	https://www.geodata.cn/main/face_science_detail?guid=34,595,274,939,620
Soil data (Clay, Sand, Silt)	1000 m	https://www.ncdc.ac.cn/portal/metadata/1fdf7dc7-7ecb-4e1f-a5df-30f7196756a8
Soil organic matter	1000 m	https://data.tpcd.ac.cn/en/data/8ba0a731-5b0b-4e2f-8b95-8b29cc3c0f3a/
Annual Average Temperature (TMP)	1000 m	https://www.geodata.cn/main/face_science_detail?guid=164304785536614
Normalized Difference Vegetation Index (NDVI)	500 m	https://www.resdc.cn/DOI/doi.aspx?DOIId=161
DEM	500 m	https://www.resdc.cn/data.aspx?DATAID=123
Human footprint (HFP)	1000 m	(Mu et al. 2022)
Socio-economic statistical data	—	China County Statistical Yearbook; social statistical bulletins; For partial missing values in the dataset, linear interpolation was adopted for imputation.

the annual mean showing periodic fluctuations accompanied by a slight declining trend. From 2001 to 2003, HNC increased amid fluctuations, followed by a fluctuating decline during 2003–2011. An improvement trend was observed during 2011–2016, whereas HNC exhibited a fluctuating downward trajectory after 2016.

The spatial distributions of HNC are presented in Figure 5a–c. Overall, HNC exhibited a distinct north–south differentiation pattern, with higher values in the south than in the north, and relatively higher levels in the southeastern and southern regions compared with NC, MYW, and NW. SC, SW, and MYG were dominated by coordinated development categories, while EC generally exhibited intermediate levels. In contrast, NC, MYW, and NW were mainly characterized by transitional development and primary coordination development categories. Regions with coordinated development displayed continuous spatial distributions in the southeastern and southern areas, whereas uncoordinated development areas accounted for a relatively small proportion and were mainly distributed as scattered patches.

In 2001, coordinated development categories accounted for 68.6% of the total area, among which moderate coordination development and high coordination development categories together accounted for 39.3%. By 2022, the proportion of coordinated development categories remained relatively stable at 67.7%, with moderate coordination development and high coordination development categories accounting for 38.4%. Specifically, the proportion of moderate coordination development category increased from 24.2% to 27.4%, whereas that of high coordination development category decreased from 15.0% to 11.0%.

Figure 5d illustrates the spatial pattern of HNC changes from 2001 to 2022. HNC showed an overall improving trend in NE and SW, whereas MYW exhibited an interspersed pattern of improvement

and degradation. In contrast, NC, MYG, and EC were dominated by declining trends, while NW and SC also showed relatively high proportions of degraded areas. Overall, 45.0% of the study area experienced improvement, whereas 54.7% underwent degradation.

Figure 5e presents the LISA spatial distribution of HNC change rates. Significant high–high clustering patterns were observed in NE and SW, with high–high clusters accounting for 59.8% and 48.5% of the respective regions. MYW and NW also contained localized high–high clustering areas. Conversely, NC and MYG exhibited prominent low–low clustering patterns, with low–low clusters accounting for 74.2% and 54.2%, respectively. Low–low clusters were also identified in EC and parts of NW. The proportions of high–low and low–high clustering types were relatively limited, displaying scattered spatial distributions mainly concentrated in transitional zones between different clustering regions.

3.3 | Temporal Development Patterns

Using Mfuzz clustering analysis, the HNC development patterns in China are classified into four types, with the following characteristics and spatial distributions (Figure 6).

Type 1 was characterized by consistently high HNC values accompanied by pronounced interannual fluctuations. This type included 611 county-level units, accounting for 21.1% of all samples. Spatially, it was mainly distributed in SC and MYG, which together accounted for 68.1% of this type, with additional distributions observed in EC and SW. Temporally, HNC remained at a relatively high level during 2001–2016, with periodic peaks occurring around 2016. After 2017, the mean HNC value declined slightly, indicating a fluctuating

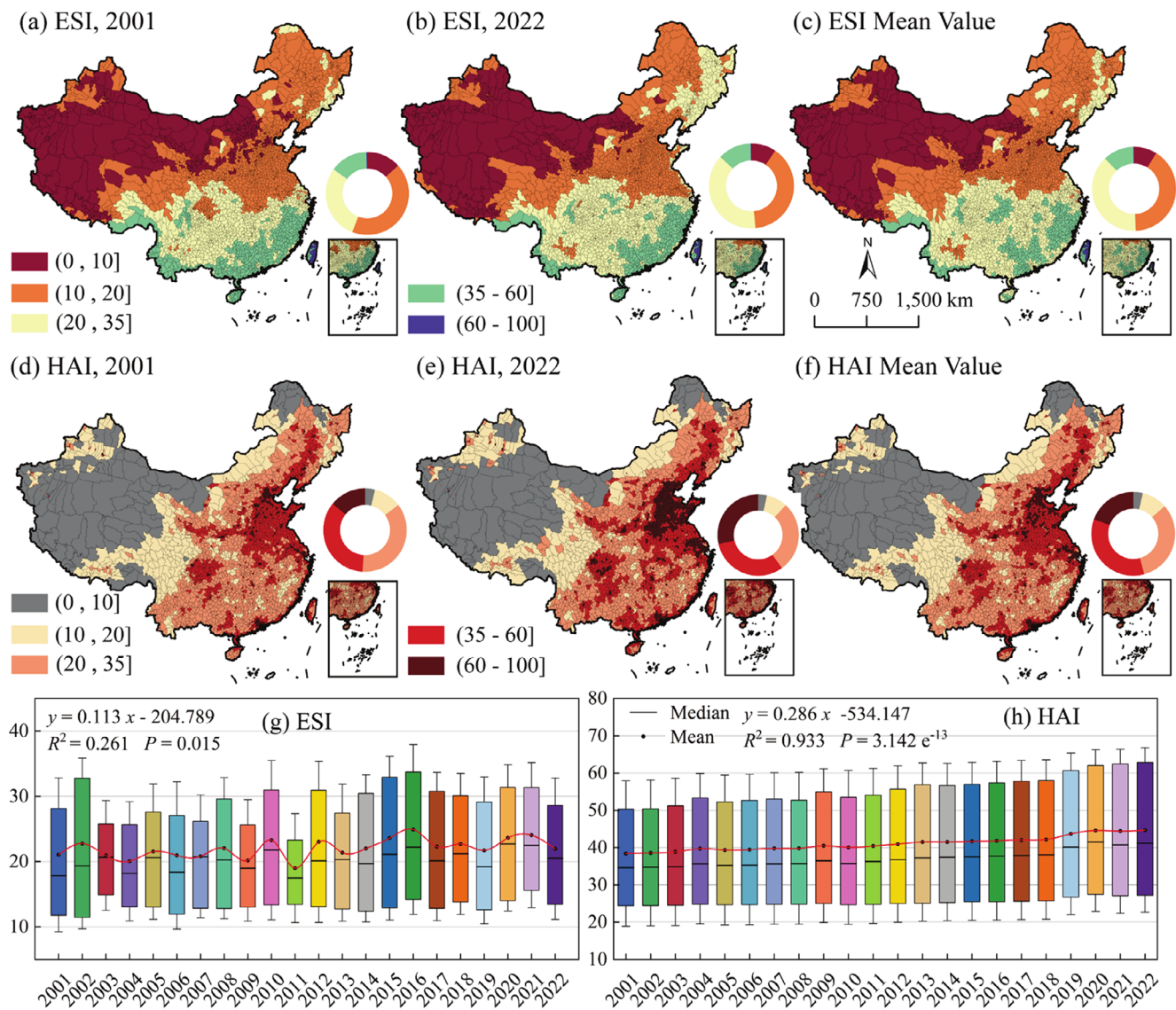


FIGURE 4 | Spatiotemporal dynamics of ESI and HAI. Wiley acknowledges that the borders within the figure are subject to multiple territorial claims. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.70834)]

adjustment process following the maintenance of a relatively high coordination level.

Type 2 exhibited a pattern of gradual improvement during the early and middle periods, followed by a decline in the later period. This type comprised 592 county-level units, representing 20.4% of all samples, and was highly concentrated in SW and MYG, which together accounted for 90.9% of this category. HNC generally increased from 2001 to approximately 2015–2016, after which it showed a declining trend. This pattern suggests that these regions experienced rapid improvements in HNC during the early period, whereas the momentum of further improvement weakened in the later stage.

Type 3 was characterized by a gradual increase in HNC from a relatively low initial level, followed by a stabilization trend. This type contained 721 county-level units, accounting for 24.9% of all samples, and was mainly distributed in NE, NW, and MYW. These areas initially exhibited relatively low HNC levels but experienced a slow upward trajectory with moderate fluctuations

throughout the study period. After 2016, the magnitude of change became relatively limited, suggesting that these regions may have entered a stage of gradual improvement in HNC.

Type 4 was characterized by persistently low HNC levels accompanied by a fluctuating downward trend. This type included 975 county-level units, accounting for 33.6% of all samples, making it the largest category among the four types. Spatially, it was primarily distributed in MYW and NC, with additional clustering observed in NW and MYG. HNC remained relatively low throughout the study period, and the low-value characteristics became more pronounced during the later stage, indicating that these regions may face persistent constraints and pressures in achieving coordinated human–nature development.

3.4 | Interpretable Analysis of MTGNN

Model performance was evaluated using MAE, MSE, RMSE, and R^2 (Table 4). The results indicate that the model achieves

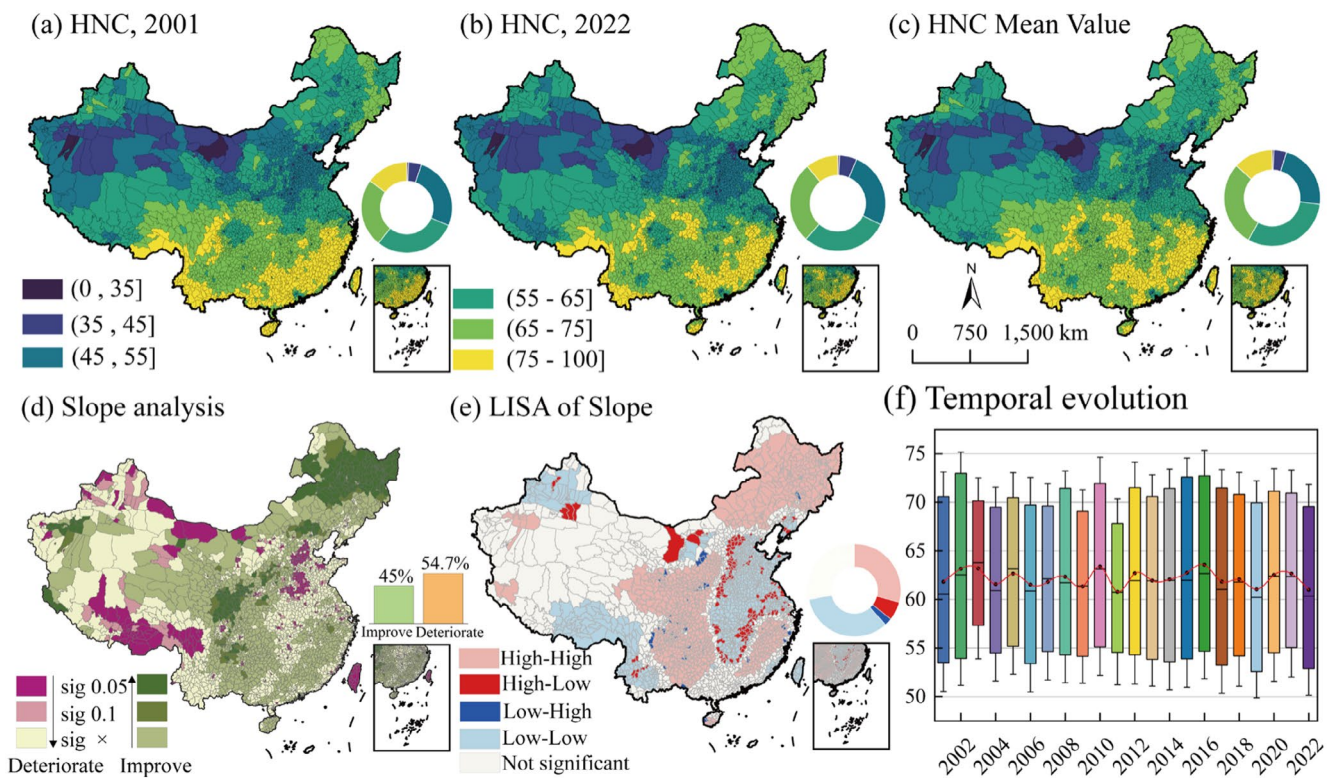


FIGURE 5 | Spatiotemporal dynamics of HNC. Wiley acknowledges that the borders within the figure are subject to multiple territorial claims. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

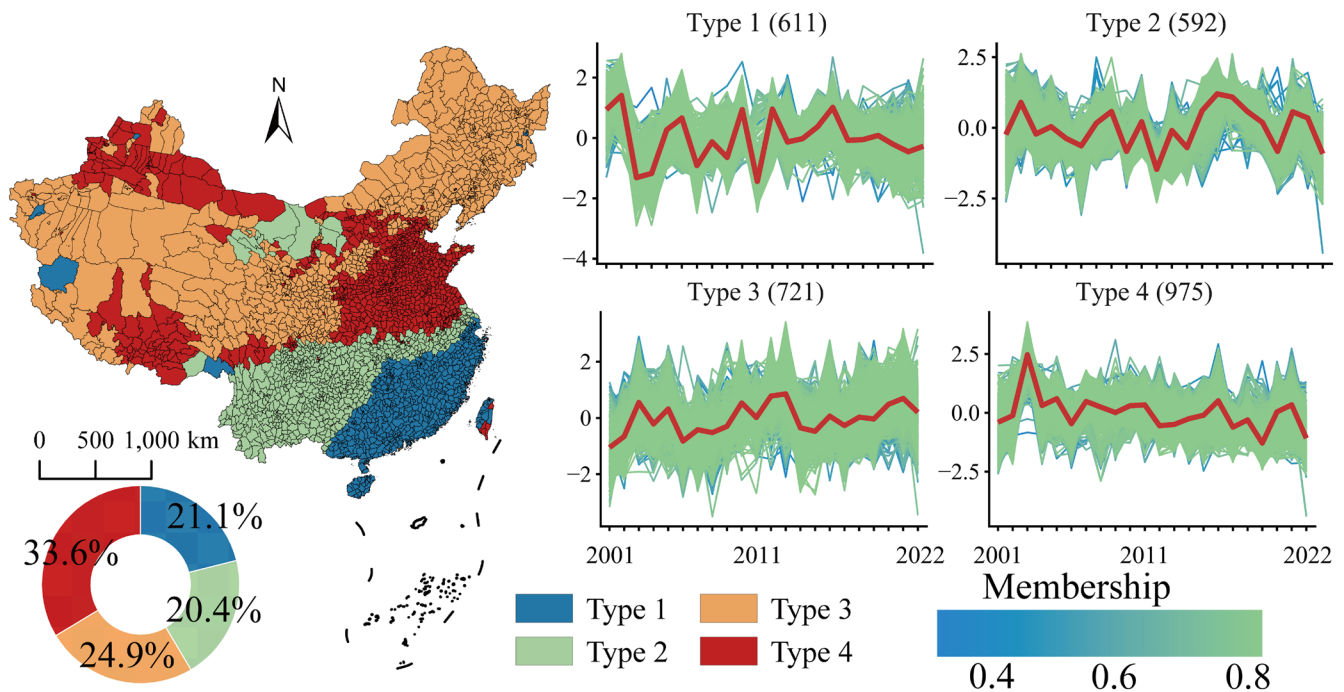


FIGURE 6 | Temporal evolution patterns of HNC. Wiley acknowledges that the borders within the figure are subject to multiple territorial claims. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

R^2 values above 0.96 in the training, validation, and test sets, demonstrating strong predictive performance. Although MAE, MSE, and RMSE are slightly higher in the test set compared with the training and validation sets, no severe overfitting is observed, and model performance remains within an acceptable range for the study's purposes.

3.4.1 | Feature Importance

The IG-based interpretability analysis of MTGNN revealed that PRE was the most influential variable in predicting HNC, with the highest relative contribution of 34.7% (Figure 7a). Sensitivity of predictive accuracy and IG attribution to alternative HNC

weighting schemes is reported in Appendix S7. TMP and SLOPE ranked second and third, with relative contributions of 12.3% and 10.3%, respectively, indicating that the model relied strongly on hydrothermal conditions and topographic characteristics when predicting HNC. The relative contributions of ISP, NDVI, and CLP were 6.6%, 5.9%, and 5.6%, respectively, suggesting that built-up land expansion, vegetation conditions, and cropland-related landscape composition also provided important predictive information for HNC.

The remaining variables exhibited relatively lower contributions, including AI (3.9%), REP (3.6%), TFA (3.2%), LFE (2.9%), SHDI (2.6%), SHAPE (2.5%), PRS (2.2%), DEM (2.0%), and SHEI (1.8%). These results demonstrate that MTGNN integrates multidimensional information from climatic conditions, topography, vegetation dynamics, land-use characteristics, landscape patterns, and socioeconomic factors to predict HNC.

Furthermore, we compared the relative contributions of different variables across temporal steps (Figure 7c). Overall, the majority of variables exhibited higher contributions at the current

time step than at lagged time steps, indicating that MTGNN primarily relies on contemporaneous information for HNC prediction. Among these variables, PRE, TMP, and SLOPE showed the highest contributions at the current time step. Meanwhile, PRE and TMP maintained relatively high contributions at the first lag period, suggesting that lagged hydrothermal conditions were also incorporated by the model when predicting current HNC. In contrast, most landscape pattern and socioeconomic variables exhibited relatively limited lagged contributions, indicating that their predictive information was mainly associated with their current states.

3.4.2 | Nonlinear Response Relationships

This study further employed partial dependence plots to examine the average response relationships between individual factors and the model-predicted HNC values (Figure 8).

PRE, SLOPE, NDVI, and CLP exhibited overall positive predictive responses, indicating that the predicted HNC values generally increased with increasing values of these variables. Among them, PRE showed the largest response magnitude, suggesting that precipitation variations corresponded to a strong response intensity in the model prediction function. SLOPE, NDVI, and CLP also demonstrated relatively consistent increasing trends, indicating that topographic gradients, vegetation conditions, and cropland-related landscape composition provided positive predictive information for HNC. PRS, DEM, SHDI, AI, and SHAPE also exhibited overall positive predictive responses, although their response magnitudes were relatively weaker.

TABLE 4 | Model performance of MTGNN.

Dataset	MAE	MSE	RMSE	R ²
Train (70%)	0.797	1.164	1.079	0.991
Validation (15%)	1.193	2.729	1.652	0.980
Test (15%)	1.564	4.906	2.215	0.961

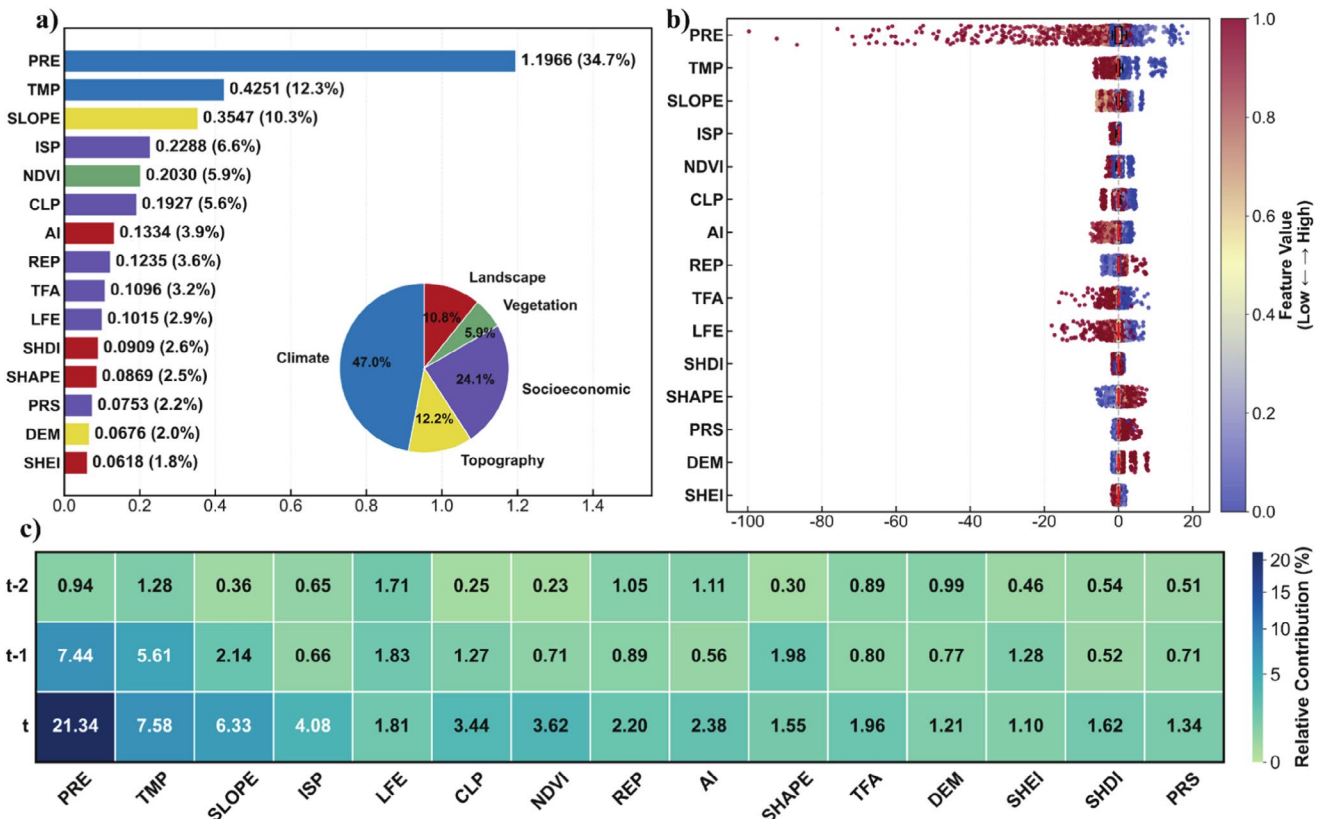


FIGURE 7 | MTGNN-based feature importance and temporal contributions. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.70834)]

In contrast, TMP, ISP, TFA, REP, LFE, and SHEI showed overall negative predictive responses, with increasing values of these variables associated with declining model-predicted HNC values. Among these factors, TMP and ISP exhibited relatively stronger response amplitudes, indicating that temperature and impervious surface proportion corresponded to more pronounced negative predictive information within the MTGNN prediction function. The negative responses of TFA, REP, LFE, and SHEI were comparatively moderate but still exhibited consistent decreasing patterns.

3.4.3 | Spatial Distribution of Dominant Predictive Factors

Figure 9 illustrates the spatial distribution of dominant IG factors in county-level HNC prediction. The dominant factor was

identified based on the maximum mean absolute IG contribution of each variable across different time steps. PRE was the dominant predictive factor in 1201 counties, accounting for 66.1% of all study units, and exhibited the most extensive distribution across China. Counties dominated by PRE were continuously distributed in SC, MYG, MYW, and NC, whereas the dominant factor combinations in SW and NW showed greater diversity.

SLOPE and TMP were identified as the dominant factors in 215 and 213 counties, respectively. Both factors exhibited strong spatial clustering characteristics. Counties dominated by SLOPE were mainly concentrated in SW, with several relatively clustered patches also observed in EC and MYW, indicating that slope-related

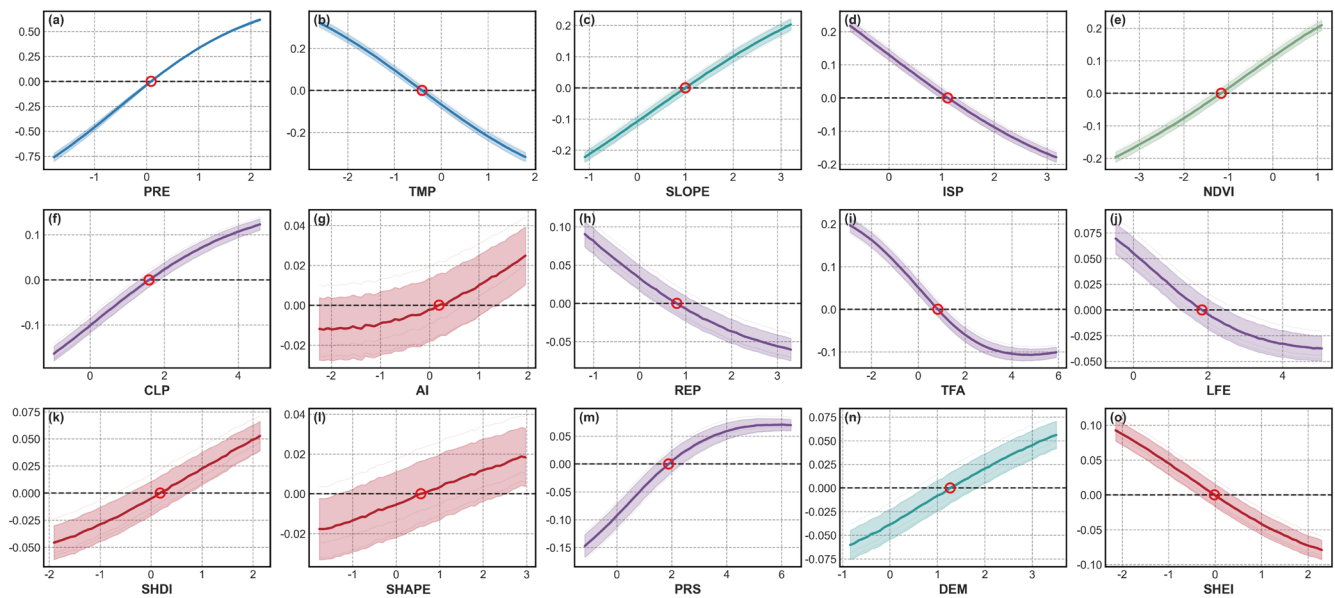


FIGURE 8 | Nonlinear response relationships. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.70834)]

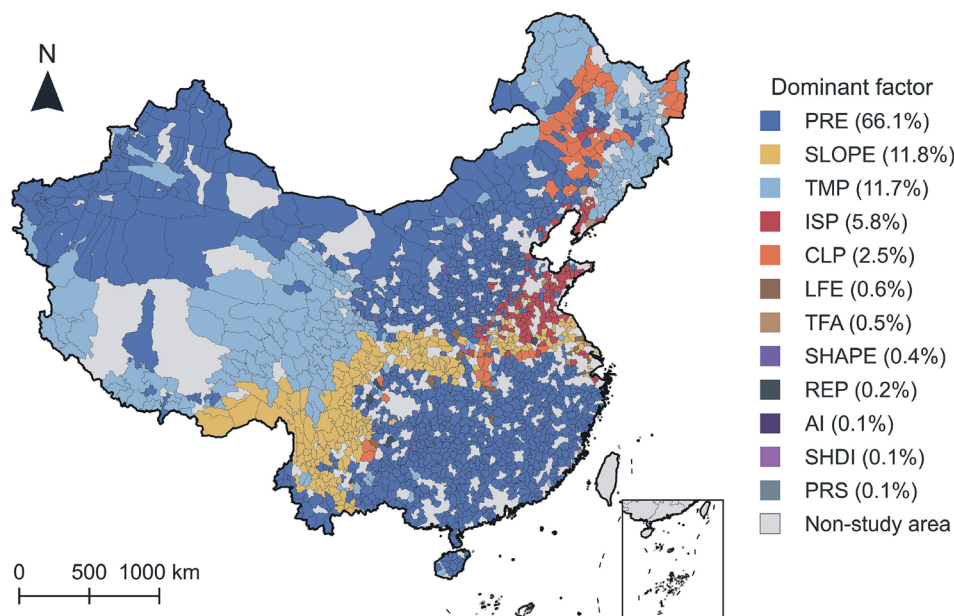


FIGURE 9 | Spatial distribution of dominant predictors. Wiley acknowledges that the borders within the figure are subject to multiple territorial claims. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.70834)]

information played a more important role in model prediction in these regions. Counties dominated by TMP were primarily distributed in NE and NW.

The dominant distributions of land-use-related factors further revealed regional heterogeneity. EC exhibited a mixed pattern dominated by PRE, SLOPE, and ISP, with ISP-dominated counties mainly concentrated in Shandong, Henan, and Jiangsu provinces. Although CLP was the dominant factor in only 46 counties, it showed a relatively concentrated distribution in NE. Other factors, including LFE, TFA, SHAPE, REP, AI, SHDI, and PRS, collectively accounted for only approximately 2.0% of the study area and were mainly characterized by scattered distributions.

3.4.4 | Learned Graph Structure of MTGNN

Figure 10 illustrates the learned county-level association structure and community characteristics derived from MTGNN. As shown in Figure 10a, high-weight connections exhibited a pronounced heterogeneous spatial distribution across China, forming an interconnected network spanning multiple regions. High-weight linkages were relatively dense in central, eastern, and southern China, whereas connections in northwestern regions and areas surrounding the Qinghai-Tibet Plateau were relatively sparse. Nevertheless, several long-distance connections with other regions were still identified. These results indicate that the county-level associations learned by the model during HNC prediction exhibited significant non-local characteristics, suggesting that information propagation among counties was not solely dependent on geographical proximity.

Based on the Louvain algorithm, four county-level communities were identified, accounting for 26.7%, 25.9%, 24.3%, and 23.0% of all counties, respectively. The four communities showed relatively balanced sizes. C1 exhibited slightly higher topographic characteristics than the overall average, while climatic, vegetation, land-use, and socioeconomic dimensions were generally close to or slightly below the average level, representing a county type primarily characterized by topographic conditions. C2 showed relatively higher values in climatic and vegetation dimensions but lower topographic characteristics, reflecting a

dominant climate-vegetation feature. C3 also exhibited relatively prominent topographic characteristics; however, it was distinguished from C1 by its relatively lower socioeconomic dimension, representing a combination of favorable terrain conditions and weaker socioeconomic development characteristics. C4 exhibited the highest socioeconomic dimension, with land-use and landscape pattern dimensions also slightly above the average, whereas vegetation and topographic dimensions were relatively lower, indicating stronger associations with human activity intensity.

From the perspective of spatial organization (Figure 10b), pronounced community interweaving characteristics were observed in densely distributed county regions across central and eastern China. In contrast, areas with larger county units in western China exhibited relatively continuous community patches. This spatial configuration suggests that the county-level association types identified from the learned graph structure showed certain degrees of spatial aggregation while retaining substantial cross-regional heterogeneity.

3.5 | Policy Effects of NKEFZ

3.5.1 | Baseline Regression

Table 5 reports the baseline regression results of the impact of NKEFZ on standardized HNC. Without incorporating fixed effects, the estimated coefficient of NKEFZ was 0.097 ($p < 0.01$). After sequentially introducing county fixed effects, county and year two-way fixed effects, and control variables, the policy coefficients were estimated as 0.096, 0.034, and 0.034, respectively, all remaining statistically significant at the 1% level. The R^2 increased substantially from 0.017 in column (1) to 0.763 in column (4), indicating that after controlling for county-specific heterogeneity, common time shocks, and climatic variations, the positive effect of NKEFZ remained robust.

3.5.2 | Parallel Trend Test

To mitigate identification bias arising from the inclusion of future-treated units as controls in the multi-period DID

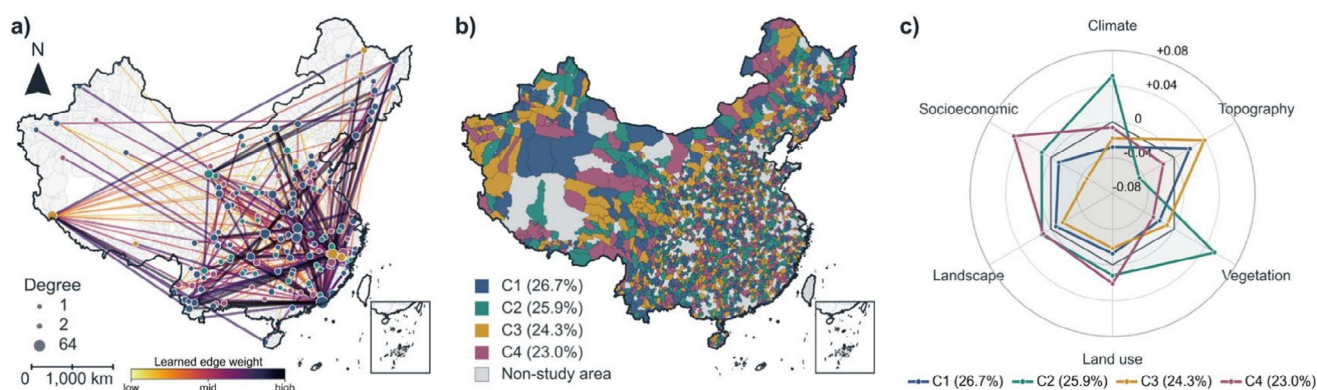


FIGURE 10 | MTGNN learned graph structure and county communities. High-weight edges represent model-learned predictive dependencies; communities summarize county groups with similar learned connections and multidimensional profiles. Wiley acknowledges that the borders within the figure are subject to multiple territorial claims. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.70834)]

framework, we used only counties that were never included in NKEFZ as the control group and constructed three event-study specifications accordingly. Experiment A considered counties incorporated into NKEFZ in 2011 as the treatment group, Experiment B considered newly incorporated counties in 2016 as the treatment group, and Experiment C combined the two groups of treated counties into a unified treatment group. These specifications examine pre-treatment trend comparability from the perspectives of initial inclusion, subsequent inclusion, and

overall policy implementation, respectively, and further identify the dynamic effects of NKEFZ on HNC.

Figure 11 presents the event-study results and CSDID estimates from the three experiments. Before policy implementation, the estimated coefficients fluctuated around zero, and the confidence intervals generally overlapped with zero, indicating no significant pre-treatment differences between treated and never-treated counties. After policy implementation,

TABLE 5 | Results of baseline regression.

	(1)	(2)	(3)	(4)
	HNC	HNC	HNC	HNC
NKEFZ	0.097***	0.096***	0.034***	0.034***
Constant	−0.015	−0.015***	−0.114***	−0.066***
Individual fixed effects	No	Yes	Yes	Yes
Time fixed effects	No	No	Yes	Yes
Control variables	No	No	No	Yes
R ²	0.017	0.017	0.181	0.763

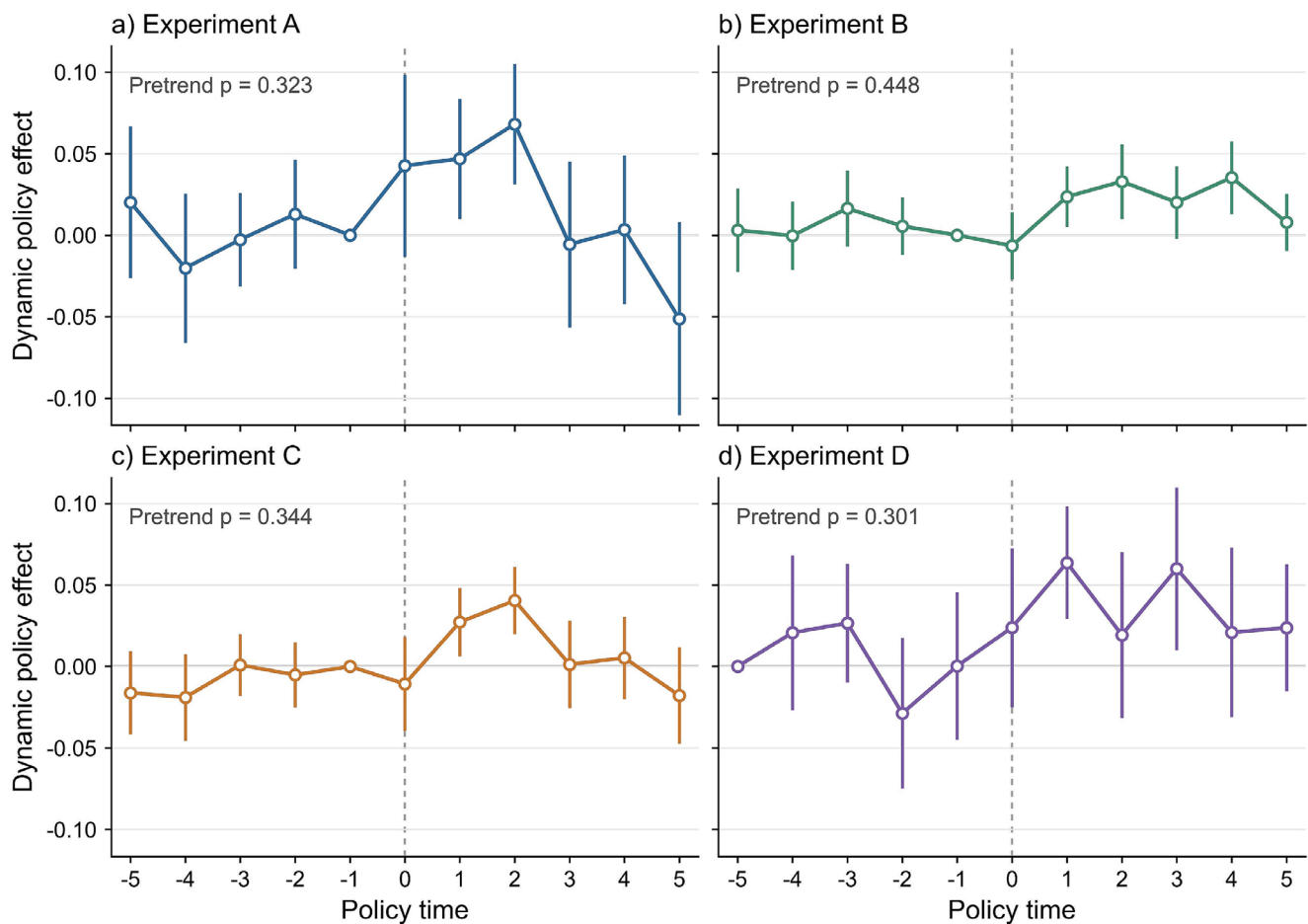


FIGURE 11 | Dynamic effects of NKEFZ. (a) 2011 cohort versus never-treated counties; (b) 2016 cohort versus never-treated counties; (c) pooled 2011 and 2016 cohorts versus never-treated counties; (d) CSDID estimates. Points indicate event-time effects and vertical bars indicate 95% confidence intervals. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Experiment A showed significantly positive effects in the first and second periods after treatment; Experiment B exhibited significant positive effects in multiple post-treatment years; and Experiment C demonstrated a clear positive shift in the early post-policy period. The consistent results across the three specifications support the comparability of pre-policy trends between treated and never-treated counties and suggest that the policy effects exhibited a certain degree of persistence.

Furthermore, this study employed the Callaway and Sant'Anna Difference-in-Differences (CSDID) approach as an additional robustness test for the multi-period DID framework. This method estimates group-time average treatment effects and provides a more flexible approach for addressing treatment effect heterogeneity under staggered adoption. It also reduces the potential "bad control group" problem caused by heterogeneous treatment timing in conventional two-way fixed-effects models, resulting in Experiment D. The results show that the *p*-value of the joint pre-treatment test in Experiment D was 0.301, indicating that the parallel trends assumption could not be rejected. Following policy implementation, the dynamic effects became consistently positive, with statistically significant effects observed in the first and third periods after treatment. These findings are consistent with the event-study results from Experiments A–C, further demonstrating that the positive effect of NKEFZ on HNC remains robust after excluding future-treated counties from the control group and accounting for treatment-effect heterogeneity in multi-period DID settings.

3.5.3 | Robustness Tests

To examine the robustness of the baseline regression results, this study conducted further analyses from four perspectives: alternative measurements of the dependent variable, different sample period specifications, placebo tests, and controls for concurrent policies (Figure 12).

Alternative measurements of the dependent variable. Taking the baseline HNC specification as the reference (R1), we further employed standardized logarithmic HNC (R2), Min-Max normalized HNC (R3), and alternative HNC measures recalculated under different weighting schemes between local and remote coordination components (R4–R7). As shown in Figure 12a, the estimated coefficients of NKEFZ remained significantly positive under all specifications, indicating that the policy effects were not sensitive to a specific normalization approach or to the weighting configuration between local and remote components.

Alternative sample period specifications. We further adjusted the sample periods by excluding observations from the policy implementation year (S1), removing samples from 2020 to 2022 (S2), and restricting the study period to before 2017 (S3). Figure 12b shows that the estimated coefficients of NKEFZ remained significantly positive across all three sample specifications. Furthermore, we defined counties incorporated into NKEFZ in 2011 (C1), counties newly incorporated in 2016 (C2), and the combined treatment group from both policy batches (C3) as treated units, with never-treated counties serving as the control group. The estimated effects remained consistently positive.

Considering that treatment assignment may be influenced by counties' initial conditions, we further conducted matched-sample tests. M1 constructed matched samples between the two policy cohorts and never-treated counties, whereas M2 implemented a propensity score matching DID approach based on pre-treatment covariates. The estimated coefficients from M1 and M2 were 0.014 and 0.018, respectively, and both were statistically significant, indicating that the baseline conclusions remained valid after improving the comparability between treated and control groups.

Placebo tests. We artificially advanced the policy implementation year for the 2016 newly incorporated counties to 2015, 2014, and 2013 (P1–P3), and similarly shifted the initial 2011 policy implementation year to 2010, 2009, and 2008 (P4–P6). Figure 12c shows that none of the placebo policy timings produced statistically significant coefficients, and all 95% confidence intervals covered zero. These results indicate that artificially assigning earlier policy timings could not reproduce the positive effects identified in the baseline regression.

Controlling for concurrent policies. We further controlled for six concurrent policy interventions, including ecological civilization demonstration zones (T1), national new-type urbanization pilot programs (T2), poverty alleviation policies (T3), rural collective property rights system reform (T4), digital village pilot programs (T5), and agricultural sustainable development pilot demonstration zones (T6). All six policies were simultaneously included in the same model in specification T7. Figure 12d demonstrates that the estimated coefficients of NKEFZ remained significantly positive after controlling for these concurrent policies individually or jointly. Overall, the robustness tests consistently yielded similar results, further supporting the conclusion that NKEFZ promotes improvements in HNC.

3.5.4 | Potential Channel Analysis

To identify the potential pathways through which NKEFZ affects HNC, this study constructed mediation models based on the baseline DID framework and further employed the Sobel test and bootstrap approach to verify the indirect effects. Table 6 shows that the direct effects remained statistically significant in the ecological restoration, industrial activity pressure, and public service support models, indicating that these pathways played partial mediating roles.

Ecological restoration pathway. The estimated coefficient of NKEFZ on NDVI was 0.065, while the coefficient of NDVI on HNC was 0.159; both were statistically significant at the 1% level. The indirect effect was estimated at 0.0104, with a Sobel test *p*-value below 0.001 and a bootstrap confidence interval of [0.0080, 0.0134]. This pathway accounted for approximately 30.9% of the total effect, indicating that vegetation restoration represents an important mechanism through which NKEFZ promotes improvements in HNC.

Industrial activity pressure pathway. The estimated coefficient of NKEFZ on AVSI was −0.109, and that of AVSI on HNC was −0.013, with both coefficients significant at the 1% level. The two negative pathways jointly generated a positive indirect

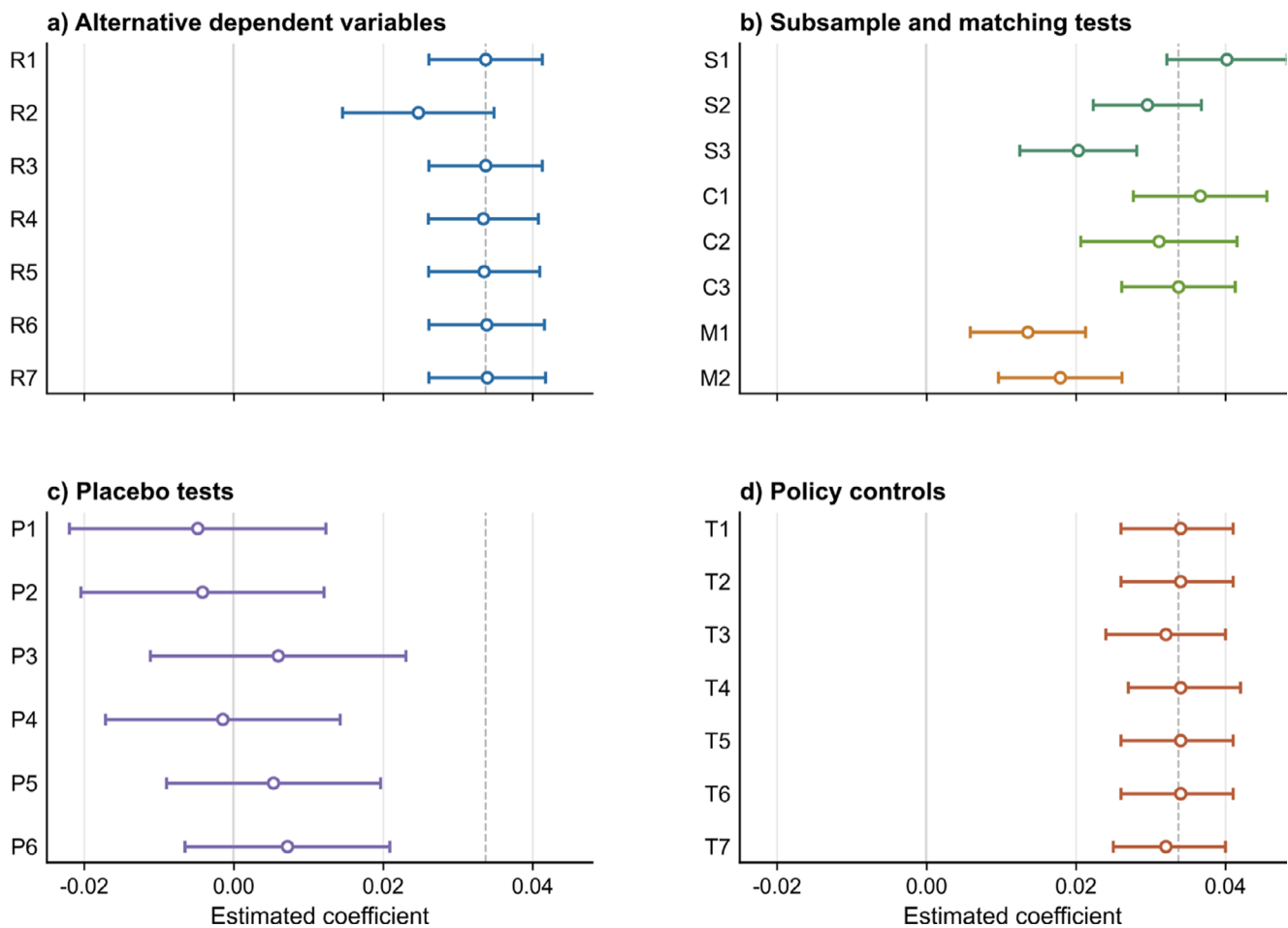


FIGURE 12 | Robustness tests for the effect of NKEFZ on HNC. (a) Alternative dependent variable tests. R1 represents the baseline standardized HNC measure. R2 and R3 use the standardized logarithmic HNC and standardized Min–Max normalized HNC, respectively. R4–R7 recalculate HNC by adjusting the weights of its two constituent dimensions to 0.35/0.65, 0.40/0.60, 0.60/0.40, and 0.65/0.35, respectively. (b) Sample adjustment and matching tests. S1–S3 correspond to excluding the policy implementation year, excluding observations from 2020 to 2022, and restricting the sample period to before 2017, respectively. C1–C3 define counties incorporated into NKEFZ in 2011, counties newly incorporated in 2016, and the combined sample of both policy cohorts as treatment groups, respectively, with counties never included in NKEFZ serving as the control group. M1 and M2 represent matched-sample DID and PSM-DID tests, respectively. (c) Placebo tests with pseudo policy timings. P1–P3 advance the policy implementation timing for counties newly incorporated in 2016 to 2015, 2014, and 2013, respectively, whereas P4–P6 shift the policy timing for counties incorporated in 2011 to 2010, 2009, and 2008, respectively. (d) Tests excluding interference from concurrent policies. T1–T6 control for ecological civilization demonstration zones, national new-type urbanization pilot programs, poverty county policies, rural collective property rights system reform, digital village pilot programs, and agricultural sustainable development pilot demonstration zones, respectively. T7 simultaneously controls for all six concurrent policies. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

effect of 0.0014, which was supported by both the Sobel and bootstrap tests. These results suggest that NKEFZ contributes to HNC improvement by constraining industrial expansion and alleviating pressures associated with resource consumption and ecological disturbance.

Industrial structure adjustment pathway. The estimated coefficient of NKEFZ on ADV was -0.019 and did not reach statistical significance. The corresponding indirect effect was -0.0001 , with a Sobel test p -value of 0.572 and a bootstrap confidence interval containing zero. Therefore, no significant transmission effect through industrial structure adjustment was identified. This mechanism may require a longer period to become fully observable.

Public service support pathway. The estimated coefficient of NKEFZ on PFE was 0.107, and that of PFE on HNC was 0.018;

both were statistically significant at the 1% level. The indirect effect was 0.0019, with a Sobel test p -value below 0.001 and a bootstrap confidence interval of [0.0008, 0.0032]. These results indicate that enhanced public service capacity provides support for county-level socioeconomic development under ecological protection constraints and constitutes another important pathway through which NKEFZ improves HNC.

3.5.5 | Heterogeneity Analysis

Regional subgroup estimates of the NKEFZ effect varied in magnitude and statistical significance across the eight regions (Figure 13a). EC exhibited the largest estimated coefficient, reaching 0.109. The coefficients for MYG, NC, and MYW were 0.059, 0.054, and 0.043, respectively, all statistically significant at the

1% level. The coefficients for NW and SW were 0.009 and 0.006, respectively, with significance only at the 10% marginal level. In contrast, the coefficients for NE and SC were 0.001 and -0.003 , respectively, none of which reached statistical significance. These regional estimates reflect the average policy responses under different sample compositions and regional development contexts.

The policy coefficients across the four MTGNN-learned graph communities ranged from 0.031 to 0.039, all statistically significant at the 1% level (Figure 13). C2 exhibited the strongest policy effect, followed by C3, with a coefficient of 0.034. C1 and C4 both showed coefficients of 0.031, indicating relatively weaker policy responses.

Furthermore, counties were grouped according to the dominant limiting-factor categories identified by MTGNN-IG (Figure 13c). To avoid an overly fragmented comparison across the 15 individual drivers, the dominant limiting factor identified for each county by MTGNN-IG was mapped to one of five broader dimensions: climate, vegetation, topography, landscape pattern, and socioeconomic conditions. The estimated coefficients were 0.051 for the socioeconomic category, 0.040 for the topographic and landscape categories, and 0.029 for the climate

category; all were statistically significant at the 1% level. The estimated coefficient for the vegetation category was not statistically significant.

4 | Discussion

4.1 | Spatiotemporal Dynamics and Driving Mechanisms of HNC

During the study period, ESI exhibited an overall fluctuating increasing trend, whereas HAI increased at a faster rate (Figure 4). Meanwhile, HNC showed a fluctuating decline after 2016 (Figure 5f). This combination suggests that the enhancement of ecosystem service supply was insufficient to continuously offset the pressures induced by expanding human activities. County-level HNC was determined not only by the level of ecological supply but also by the magnitude and growth rate of human activities. Therefore, assessments of HNC should simultaneously consider the changing directions and relative rates of the two subsystems, highlighting the dynamic, multidimensional, and context-dependent characteristics of human–nature relationships (Ellis et al. 2025; Wang et al. 2025).

TABLE 6 | Results of potential channel analysis.

Potential channel	Mediator	NKEFZ → channel	Channel → HNC	Direct effect	Indirect effect	Sobel <i>p</i> -value	Bootstrap 95% CI
Ecological restoration	NDVI	0.065***	0.159***	0.023***	0.0104***	<0.001	[0.0080, 0.0134]
Industrial activity pressure	AVSI	−0.109***	−0.013***	0.032***	0.0014***	<0.001	[0.0007, 0.0022]
Industrial structure adjustment	ADV	−0.019	0.003**	0.034***	−0.0001	0.572	[−0.0003, 0.0002]
Public service support	PFE	0.107***	0.018***	0.032***	0.0019***	<0.001	[0.0008, 0.0032]

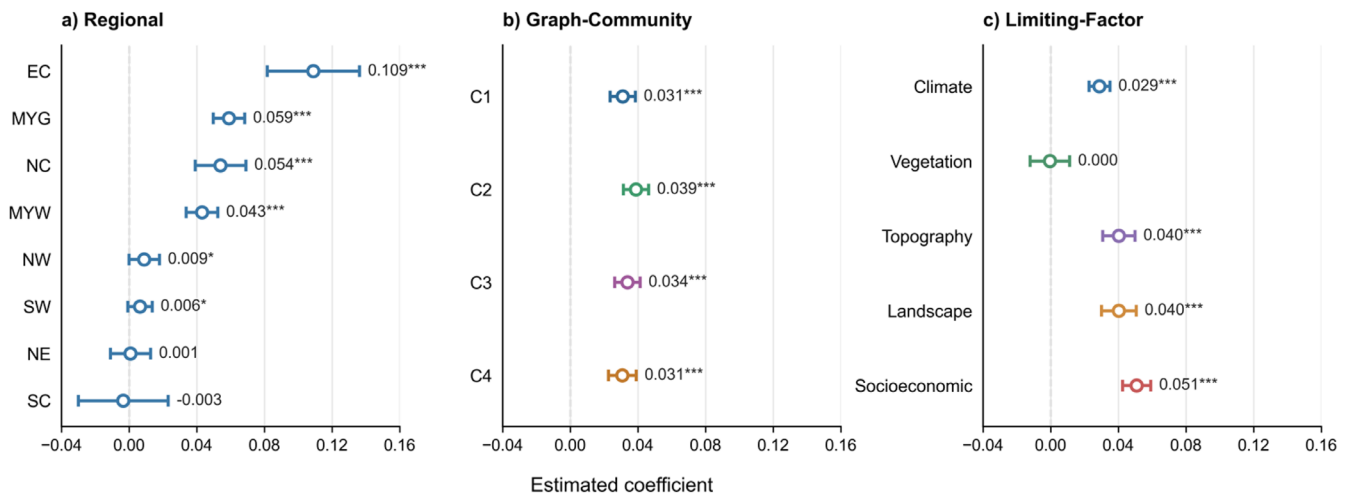


FIGURE 13 | Heterogeneity analysis. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

HNC exhibited a clear spatial differentiation pattern across China (Figure 5a–e), with generally higher values observed in southern and southeastern counties, although some areas experienced declines during the later stage of the study period. NE and SW initially exhibited relatively lower HNC levels but developed extensive areas with improvement trends. The four temporal evolution trajectories (Figure 6) further demonstrated that counties with initially high HNC levels may experience subsequent declines, whereas counties with relatively low initial conditions may enter a gradual improvement pathway. Although the initial state of counties influences their subsequent evolution, it does not solely determine long-term trajectories. Integrating spatial levels, changing trends, and temporal trajectories provides a more comprehensive understanding of the maintenance pressures faced by high-HNC regions and the improvement potential of low-HNC regions.

The MTGNN-IG results further revealed the driving mechanisms underlying HNC spatiotemporal dynamics (Table 4; Figure 7). PRE and TMP together accounted for 47.0% of the relative contribution, and both maintained high contributions at the current and one-period lagged time steps, indicating that HNC was highly sensitive to hydrothermal conditions and their interannual persistence. PDP analysis (Figure 8) showed that PRE generally exhibited a positive predictive response, whereas TMP showed an overall negative predictive response, reflecting the contrasting roles of water availability and thermal stress in model predictions. The lagged contributions of climatic variables further suggest that climate effects may influence subsequent HNC development through processes related to vegetation growth, ecosystem service provision, and agricultural production (Chen et al. 2023; Li, Chen, et al. 2025).

The spatial distribution of dominant predictive factors further revealed substantial regional heterogeneity in the major constraints affecting HNC (Figure 9). PRE was identified as the dominant predictive factor in 66.1% of counties and exhibited relatively continuous distributions in SC, MYG, MYW, and NC. SLOPE and TMP dominated in 11.8% and 11.7% of counties, respectively, with the former mainly concentrated in SW and the latter primarily distributed in NE and NW. EC exhibited a mixed pattern dominated by PRE, SLOPE, and ISP. These findings indicate that even counties with similar HNC levels or changing trajectories may rely on different predictive information within the model, suggesting that analyses based on unified driving factors may overlook substantial spatial heterogeneity.

The MTGNN-learned graph further revealed county-level associations beyond geographical proximity (Figure 10). High-weight connections extended across non-adjacent counties, and the four identified communities exhibited a mosaic spatial distribution nationwide. This pattern indicates that geographically distant counties may undergo similar evolutionary processes due to shared climate responses, land-use changes, and socioeconomic dynamics. The distance-weighted remote component in HNC captures potential spatial linkages, whereas the learned graph extracts predictive dependencies from multivariate temporal sequences; together, they expand the spatial scope for understanding county-level human–nature system interactions. This finding is consistent with previous studies emphasizing teleconnections among distant systems and highlighting that

ecosystem service flows are jointly shaped by supply–demand configurations and spatial structures (Assis et al. 2023; Liu et al. 2013; Ma et al. 2024).

4.2 | Policy Effects and Potential Pathways of NKEFZ

During the study period, HAI increased faster than ESI, resulting in increasing pressure on national HNC. Against this background, NKEFZ still generated a stable positive effect, indicating that spatial regulation, ecological restoration, and fiscal support oriented toward ecological function conservation can alleviate the persistent pressures imposed by development activities on HNC. The baseline DID results (Table 5) and a series of robustness tests (Figures 11, 12) consistently support the conclusion that NKEFZ improves HNC.

The pathway analysis further revealed the potential mechanisms underlying the policy effects (Table 6). Ecological restoration strengthened the ecological foundation, reduced industrial activity pressure, alleviated resource consumption and ecological disturbance, and public service support mitigated constraints on local development capacity and residents' welfare caused by conservation responsibilities. These three pathways operate through ecological supply, development pressure, and social support capacity, respectively, jointly promoting a more balanced relationship between ESI and HAI.

Ecological restoration represented the most evident policy transmission pathway (Table 6), with the indirect effect of NDVI accounting for approximately 30.9% of the total effect. This indicates that ecological restoration, vegetation conservation, and spatial land-use regulation can be relatively rapidly translated into improved ecological conditions and subsequently enhance HNC. This finding is consistent with previous studies showing that NKEFZ contributes to vegetation improvement and ecosystem function enhancement (Li et al. 2021; Xu et al. 2025). Although the indirect effects through AVSI and PFE were relatively small, both pathways provided important complementary functions. Controlling industrial activity pressure creates additional space for ecological recovery, whereas public service support reduces the opportunity costs faced by local governments responsible for ecological conservation. Consequently, ecological improvement, pressure mitigation, and maintenance of local development capacity constitute a mutually reinforcing policy package (Wen et al. 2025).

The indirect effect of industrial structure adjustment was not statistically significant (Table 6). This result reflects differences in the temporal responses of various policy instruments. Industrial activity intensity, development boundaries, and fiscal investment can be adjusted relatively quickly through administrative regulation or financial arrangements, whereas industrial upgrading depends on enterprise capacity, technological diffusion, human capital accumulation, market demand, and employment transformation. Adjustments in industrial intensity generally precede the formation of green industries; therefore, the current benefits of NKEFZ are primarily derived from ecological restoration, development pressure control, and public service support. Future cultivation of green industries and the

realization of ecological product values may serve as complementary pathways to consolidate long-term policy effects (Fan et al. 2024).

Significant heterogeneity (Figure 13) existed across regions and dominant constraint groups, indicating that the extent to which the same policy instruments improve HNC depends on county-level initial constraints and regional development contexts. EC, MYG, NC, and MYW exhibited stronger policy responses, whereas the incremental effects in NW and SW were relatively limited, and significant effects were not detected in NE and SC. The grouping based on dominant constraints provided further insights: socioeconomic constraint counties exhibited the strongest policy responses, while terrain, landscape, and climate constraint groups also benefited positively from NKEFZ. In contrast, vegetation constraint counties showed relatively weak overall policy effects. Existing policy instruments can directly adjust fiscal support, development activities, and spatial governance conditions; however, HNC improvement in vegetation-constrained counties may additionally depend on restoration quality, ecosystem functions, and other interacting constraints.

All four MTGNN graph communities exhibited stable positive policy responses. The functional associations learned by the model identified counties with similar spatiotemporal evolutionary processes, extending the understanding of NKEFZ policy effects beyond geographically adjacent regions. This indicates that positive policy responses can emerge across diverse non-local county association structures. Policy intensity and instrument allocation should therefore be primarily determined by regional contexts and dominant constraints, while graph communities can provide additional references for comparison and policy learning among functionally similar counties.

4.3 | Policy Implications Based on an Integrated MTGNN-IG-DID Framework

The combined evidence from MTGNN-IG and DID provides insights into three interconnected governance questions: what factors primarily constrain county-level HNC, to what extent existing NKEFZ instruments generate improvements in local contexts, and which policy instruments are more likely to produce substantial benefits. Accordingly, optimization of NKEFZ implementation should follow a decision-making framework of “identifying dominant constraints, evaluating policy responses, and allocating policy instruments,” and should be implemented across three levels: regional organization, county-level governance, and cross-regional coordination.

In EC, dominant factors exhibited a combined pattern of PRE, SLOPE, and ISP, accompanied by relatively strong policy responses (Figure 9). For these counties, spatial land-use regulation and industrial access management should continue to be strengthened. Particular attention should be given to ISP-dominated counties, where controlling construction land expansion, improving industrial land-use efficiency, and integrating ecological restoration into territorial spatial governance should be prioritized. Green technological transformation and low ecological-impact industries can serve as long-term complementary strategies (Xu et al. 2025), gradually transforming

development pressure control into more sustainable development pathways.

MYG, MYW, and NC were primarily characterized by PRE as the dominant predictive factor, and NKEFZ also demonstrated relatively stable positive effects in these regions. Policy priorities should focus on water conservation, watershed ecological restoration, water-use efficiency improvement, and adaptation to drought and flood risks. In counties facing strong socioeconomic constraints, ecological compensation, public service provision, and green employment support should be coordinated to enhance local capacity for sustaining ecological conservation responsibilities (Zhu et al. 2025). Such policy combinations can simultaneously maintain ecosystem services supported by water availability while preserving the socioeconomic foundations of conservation regions.

In NE and NW, dominant factors were more frequently associated with TMP, and short-term policy responses were relatively limited. These regions should incorporate thermal risks, drought monitoring, soil moisture conservation, and vegetation resilience enhancement into NKEFZ implementation strategies. Agricultural production systems and ecological restoration approaches should be adjusted according to local water resource carrying capacity. SW was primarily constrained by SLOPE, and the incremental improvement of HNC induced by existing policies was relatively limited. Governance priorities should therefore shift toward slope-use regulation, soil and water conservation, geological hazard prevention, and low-disturbance livelihoods, while improving the suitability of public services and infrastructure in mountainous areas. Ecological engineering planning should fully consider topographic constraints to avoid conflicts between restoration projects and local production-living spaces. SC generally exhibited high HNC levels, but its average policy response remained unclear. Therefore, a combined strategy of maintaining existing advantages and implementing targeted improvements is more appropriate. Particular attention should be paid to counties experiencing HNC decline, with strengthened management of localized pressures arising from tourism development, transportation infrastructure expansion, and urban growth.

Regional classification provides a framework for organizing policy implementation, but specific instruments should still be tailored according to dominant county-level constraints. Socioeconomic constraint counties should prioritize ecological compensation, public services, and green employment support. Landscape constraint counties should emphasize ecological corridor construction, fragmentation mitigation, and ecological connectivity enhancement. Terrain constraint counties require slope management and adaptive infrastructure, whereas climate constraint counties should strengthen hydrothermal risk management and ecosystem resilience. For vegetation constraint counties, policy objectives should shift from merely expanding vegetation coverage toward improving community structure, restoration quality, and ecosystem service functions, while simultaneously addressing climatic, topographic, and socioeconomic constraints.

MTGNN graph communities can complement conventional administrative governance systems (Figure 10). Counties

belonging to the same graph community may be geographically distant but share similar predictive dependencies and HNC evolutionary processes, providing potential reference areas for policy comparison, monitoring, and knowledge exchange. Graph communities are therefore more suitable for identifying cross-regional learning relationships and functional cooperation networks, while the intensity of policy investment should continue to depend on county-specific constraints and governance requirements.

Overall, regional divisions can support the organization of policy implementation and fiscal allocation; dominant constraints can guide the selection of county-level policy instruments, and graph communities provide connections for policy learning among non-adjacent counties. Ecological restoration, industrial activity pressure control, and public service support constitute fundamental policy tools, whereas their specific allocation should be adjusted according to regional environments, county-level constraints, and policy responsiveness. This multi-level governance framework can translate national ecological conservation objectives into differentiated county-level implementation strategies.

4.4 | Research Boundaries and Future Directions

This study still has several aspects that warrant further investigation. First, the current analysis identifies the average county-level effects of the NKEFZ policy package but provides limited insight into the implementation intensity of individual policy instruments and responses at the micro level. Different instruments, including ecological transfer payments, restoration project investments, industrial access regulation, ecological migration, and ecological product value realization, may operate through distinct temporal pathways. Future studies could integrate detailed fiscal allocation data, project implementation records, household surveys, and ecological product transaction data to further identify how specific policy instruments influence local conservation capacity, residents' welfare, and green industrial development.

Second, the MTGNN-learned graph characterizes functional associations among counties based on multivariate temporal sequences, whereas the actual mechanisms underlying these associations could be further enriched by incorporating real regional flow data (Assis et al. 2023; Ma et al. 2024). Population migration, transportation connectivity, commodity trade, water transfers, pollution diffusion, and ecosystem service flows may all shape human–nature system interactions among counties. Future research could integrate these flow data into multi-layer graph structures, compare the consistency between learned predictive associations and observed interaction networks, and examine the relative contributions of different cross-regional connections to HNC dynamics.

Third, this study identifies the evolutionary characteristics, predictive factors, and policy effects of HNC based on historical observations from 2001 to 2022 but does not simulate long-term changes under alternative development scenarios. Future research could incorporate climate change, land-use transformation, population mobility, urbanization, industrial restructuring,

and ecological protection intensity into scenario-based projections. Comparing long-term HNC trajectories and spatial differentiation under different development pathways would provide forward-looking evidence for optimizing NKEFZ implementation and promoting regional sustainable development.

5 | Conclusion

This study developed an integrated framework that couples an interpretable multivariate time-series graph neural network (MTGNN-IG) with a staggered difference-in-differences (DID) design to jointly characterize the dynamics, predictive constraints, and policy responsiveness of HNC across counties in China from 2001 to 2022.

Three main conclusions emerge. First, national HNC remained at the primary-coordination level but declined slightly after 2016, with 45.0% of counties improving and 54.7% deteriorating; a pronounced south-high/north-low pattern and four distinct temporal trajectories indicate that county HNC is governed jointly by the levels and the relative growth rates of ecosystem-service supply and human activity, rather than by initial conditions alone. Second, the MTGNN-IG attribution identified precipitation, temperature, and slope as the dominant predictors (34.7%, 12.3%, and 10.3% of contribution), with precipitation dominant in 66.1% of counties, and revealed spatially heterogeneous limiting factors together with non-local predictive communities that extend beyond geographical proximity. Third, NKEFZ produced a stable positive effect on HNC, robust across alternative estimators, measures, placebo, and concurrent-policy tests; ecological restoration (the NDVI pathway) accounted for approximately 30.9% of the total effect, reduced industrial-activity pressure, and public-service support acted as complementary channels, whereas industrial-structure upgrading was not statistically supported, and effects varied markedly across regions and limiting-factor groups.

By linking predictive constraints to policy responsiveness, the framework supports a constraint-sensitive governance logic—identify dominant constraints, evaluate policy responses, and allocate instruments accordingly—for differentiated county-scale sustainable development. Because the contribution reflects predictive associations and the pathways are interpreted as potential rather than strictly causal channels, future work incorporating direct cross-county flow data and scenario-based projections would further strengthen these conclusions.

Acknowledgments

This research was funded primarily by the National Natural Science Foundation of China with grant number (42071228).

Funding

This work was supported by the National Natural Science Foundation of China (42071228).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The MTGNN model architecture and analysis code used in this study are openly available at <https://github.com/Albert-007MAX/MTGNN-panel-regression-code>. Other data supporting the findings of this study are available from the corresponding author upon reasonable request.

References

- An, M., P. Xie, W. He, B. Wang, J. Huang, and R. Khanal. 2023. "Local and Tele-Coupling Development Between Carbon Emission and Ecologic Environment Quality." *Journal of Cleaner Production* 394: 136409. <https://doi.org/10.1016/j.jclepro.2023.136409>.
- Assis, J. C., C. Hohlenwerger, J. P. Metzger, et al. 2023. "Linking Landscape Structure and Ecosystem Service Flow." *Ecosystem Services* 62: 101535. <https://doi.org/10.1016/j.ecoser.2023.101535>.
- Callaway, B., and P. H. C. Sant'Anna. 2021. "Difference-In-Differences With Multiple Time Periods." *Journal of Econometrics* 225: 200–230. <https://doi.org/10.1016/j.jeconom.2020.12.001>.
- Cao, H., Y. Qi, J. Chen, S. Shao, and S. Lin. 2021. "Incentive and Coordination: Ecological Fiscal Transfers' Effects on Eco-Environmental Quality." *Environmental Impact Assessment Review* 87: 106518. <https://doi.org/10.1016/j.eiar.2020.106518>.
- Chang, D., Z. Zhang, X. Yang, and Z. Ma. 2025. "Ecological Environmental Space Control, Land Transaction Price and Green Development: A Quasi-Natural Experiment Based on National Key Ecological Function Areas." *Journal of Quantitative & Technological Economics* 42: 69–90. <https://doi.org/10.13653/j.cnki.jqte.20250113.004>.
- Chen, S., S. Yao, M. Hou, and F. Weng. 2024. "Transfer Payment in National Key Ecological Functional Areas and Supply of Basic Public Services: Empirical Analysis From Counties in China." *East China Economic Management* 38: 13–24. <https://doi.org/10.19629/j.cnki.34-1014/f.230723004>.
- Chen, Z., W. Wang, A. Cescatti, and G. Forzieri. 2023. "Climate-Driven Vegetation Greening Further Reduces Water Availability in Drylands." *Global Change Biology* 29: 1628–1647. <https://doi.org/10.1111/gcb.16561>.
- Cheng, Y., B. Ma, and Y. Sun. 2023. "Does central ecological transfer payment enhance local environmental performance? Quasi-experimental evidence from China." *Ecological Economics* 212: 107920. <https://doi.org/10.1016/j.ecolecon.2023.107920>.
- Ellis, E. C., Y. Malhi, H. Ritchie, et al. 2025. "An Aspirational Approach to Planetary Futures." *Nature* 642: 889–899. <https://doi.org/10.1038/s41586-025-09080-1>.
- Fan, D., and W. Wei. 2025. "Ecological Projects Highly Affect Ecosystem Resistance and Resilience to Drought Stress: Evidence From the Yellow River Basin in China." *Ecosystem Health and Sustainability* 11: 0371. <https://doi.org/10.34133/ehs.0371>.
- Fan, D., and F. Yang. 2024. "Assessing the Impact of China's Agricultural Subsidy Reform on Fertilizer Management: A County-Level Empirical Analysis Based on Difference-In-Difference Model." *Frontiers in Sustainable Food Systems* 7: 1298425. <https://doi.org/10.3389/fsufs.2023.1298425>.
- Fan, Q., H. Liu, Y. Liu, and Z. Wang. 2024. "Does Ecological Compensation in the Beijing-Tianjin-Hebei Region Affect Eco-Environmental Protection—Evidence From the Key Subsidies for Transfer Payment in National Key Ecological Functional Areas." *China Journal of Economics* 11: 416–444. <https://doi.org/10.26599/CJE.2024.9300115>.
- Fan, Y., X. Jin, L. Gan, et al. 2022. "Dynamics of Spatial Associations Among Multiple Land Use Functions and Their Driving Mechanisms: A Case Study of the Yangtze River Delta Region, China." *Environmental Impact Assessment Review* 97: 106858. <https://doi.org/10.1016/j.eiar.2022.106858>.
- Fu, B. 2020. "United Nations Sustainable Development Goals and the Historical Mission of Geographical Science." *Science & Technology Review* 38: 19–24.
- Gong, C., J. Zhang, and H. Liu. 2021. "Do Industrial Pollution Activities in China Respond to Ecological Fiscal Transfers? Evidence From Payments to National Key Ecological Function Zones." *Journal of Environmental Planning and Management* 64: 1184–1203. <https://doi.org/10.1080/09640568.2020.1813695>.
- Hao, H., Y. Li, H. Zhang, R. Zhai, and H. Liu. 2019. "Spatiotemporal Variations of Vegetation and Its Determinants in the National Key Ecological Function Area on Loess Plateau Between 2000 and 2015." *Ecology and Evolution* 9: 5810–5820. <https://doi.org/10.1002/ece3.5165>.
- Hu, H., K. Yan, Y. Shi, T. Lv, X. Zhang, and X. Wang. 2024. "Decrypting Resilience: The Spatiotemporal Evolution and Driving Factors of Ecological Resilience in the Yangtze River Delta Urban Agglomeration." *Environmental Impact Assessment Review* 106: 107540. <https://doi.org/10.1016/j.eiar.2024.107540>.
- Karri, R. R., G. Ravindran, V. Pingili, N. M. Mubarak, K. N. Ruslan, and Y. H. Tan. 2026. "Integrating the Food-Energy-Water Nexus: Strategies for Climate Change Mitigation With SDG Alignment." *Environmental Impact Assessment Review* 116: 108070. <https://doi.org/10.1016/j.eiar.2025.108070>.
- Kong, F., H. Chen, and S. Bao. 2026. "Analysis of the Dynamic Evolution, Regional Differences, and Influencing Factors of Green Total Factor Productivity in Grain: The Case of China." *Environmental Impact Assessment Review* 116: 108141. <https://doi.org/10.1016/j.eiar.2025.108141>.
- Li, G., L. Li, X. Li, and Y. Chen. 2021. "Can the Establishment of National Key Ecological Functional Zones Improve Air Quality?: An Empirical Study From China." *PLoS One* 16: e0246257. <https://doi.org/10.1371/journal.pone.0246257>.
- Li, R., Z. Du, Y. Yan, S. Wu, Z. Wu, and K. Ma. 2025. "Integrating Ecosystem Health and Human Footprint in Ecological Function Zoning: A Framework for Sustainable Landscape Management in Sanchuan River Watershed, China." *Landscape Ecology* 40: 213. <https://doi.org/10.1007/s10980-025-02211-z>.
- Li, R., Y. Yu, W. Cai, Y. Liu, and Y. Li. 2024. "Exploring the Gradient Impact of Climate and Economic Geographical Factors on City-Level Building Carbon Emissions in China: Characteristics and Enlightenments." *Sustainable Cities and Society* 113: 105637. <https://doi.org/10.1016/j.scs.2024.105637>.
- Li, X., Y. Chen, S. Li, X. Sun, Q. Liu, and H. Wang. 2025. "Precipitation and Soil Texture Dominate the Spatiotemporal Changes in the Carbon-Water Coupling Coordination in Taihang Mountains, China." *Journal of Geographical Sciences* 35: 521–551. <https://doi.org/10.1007/s11442-025-2333-4>.
- Liao, Z., K. Su, X. Jiang, et al. 2023. "Spatiotemporal Variation and Coupling of Grazing Intensity and Ecosystem Based on Four Quadrant Model on the Inner Mongolia." *Ecological Indicators* 152: 110379. <https://doi.org/10.1016/j.ecolind.2023.110379>.
- Liu, J., V. Hull, M. Batistella, et al. 2013. "Framing Sustainability in a Telecoupled World." *Ecology and Society* 36: 7871–7885. <https://doi.org/10.5846/stxb201611152318>.
- Liu, M., Q. Luo, R. Huang, Y. Wu, and J. Zhou. 2025. "Dynamic Impacts of Urban Expansion on Vegetation Growth in Urban Environments: A Sustainable Land Management Framework." *Land Use Policy* 153: 107541. <https://doi.org/10.1016/j.landusepol.2025.107541>.
- Liu, Q., J. Qiao, D. Han, M. Li, and L. Shi. 2023. "Spatiotemporal Evolution of Cultivated Land Use Eco-Efficiency and Its Dynamic Relationship With Landscape Pattern Change From the Perspective of Carbon Effect: A Case Study of Henan, China." *Agriculture* 13: 1350. <https://doi.org/10.3390/agriculture13071350>.

- Liu, Q., J. Qiao, M. Li, Y. Dun, X. Zhu, and X. Ji. 2024. "Spatiotemporal Evolution of Ecological Environmental Quality and Its Dynamic Relationships With Landscape Pattern in the Zhengzhou Metropolitan Area: A Perspective Based on Nonlinear Effects and Spatiotemporal Heterogeneity." *Journal of Cleaner Production* 480: 144102. <https://doi.org/10.1016/j.jclepro.2024.144102>.
- Liu, Q., J. Qiao, M. Li, and M. Huang. 2023. "Spatiotemporal Heterogeneity of Ecosystem Service Interactions and Their Drivers at Different Spatial Scales in the Yellow River Basin." *Science of the Total Environment* 908: 168486. <https://doi.org/10.1016/j.scitotenv.2023.168486>.
- Lokesh, K., and E. Matthias. 2019. "Mfuzz: A Software Package for Soft Clustering of Microarray Data." *Bioinformation* 2: 5–7. <https://doi.org/10.6026/97320630002005>.
- Lu, D. 2003. "Some Key Issues Concerning Development of Geographical Science in China." *Acta Geographica Sinica* 58: 3–8. <https://doi.org/10.11821/xb200301001>.
- Lu, W., and C. Zhang. 2023. "Does Ecological Compensation Narrow the Urban-Rural Income Gap?—Empirical Evidence Based on Transfer Payments in National Key Ecological Function Zone of China." *Public Finance Research* 44: 82–98. <https://doi.org/10.19477/j.cnki.11-1077/f.2023.07.006>.
- Ma, S., G. Deng, L.-J. Wang, H. Hu, X. Fang, and J. Jiang. 2024. "Telecoupling Between Urban Expansion and Forest Ecosystem Service Loss Through Cultivated Land Displacement: A Case Study of Zhejiang Province, China." *Journal of Environmental Management* 357: 120695. <https://doi.org/10.1016/j.jenvman.2024.120695>.
- Ministry of Housing and Urban-Rural Development of the People's Republic of China. 2025. "Statistical Bulletins and Public Data on Housing and Urban - Rural Development." <https://www.mohurd.gov.cn/gongkai/fdzdgknr/sjfb/tjxx/tjgb/index.html>.
- Mu, H., X. Li, Y. Wen, et al. 2022. "A Global Record of Annual Terrestrial Human Footprint Dataset From 2000 to 2018." *Scientific Data* 9: 176. <https://doi.org/10.1038/s41597-022-01284-8>.
- Qi, T., Q. Ren, D. Zhang, W. Lu, and C. He. 2023. "Impacts of Urban Expansion on Vegetation in Drylands: A Multiscale Analysis Based on the Vegetation Disturbance Index." *Ecological Indicators* 147: 109984. <https://doi.org/10.1016/j.ecolind.2023.109984>.
- Shen, B., X. Yi, Y. Sun, et al. 2020. "Proteomic and Metabolomic Characterization of COVID-19 Patient Sera." *Cell* 182: 59–72. <https://doi.org/10.1016/j.cell.2020.05.032>.
- Su, X., Y. Fan, and C. Wen. 2024. "Systematic Coupling and Multistage Interactive Response of the Urban Land Use Efficiency and Ecological Environment Quality." *Journal of Environmental Management* 365: 121584. <https://doi.org/10.1016/j.jenvman.2024.121584>.
- Sundararajan, M., A. Taly, and Q. Yan. 2017. "Axiomatic Attribution for Deep Networks. arXiv." <https://doi.org/10.48550/arXiv.1703.01365>.
- Wang, C., Y. Liu, J. Chen, and C. Yu. 2025. "Turning Points of the Relationship Between Human Activity and Environmental Quality in China." *Sustainable Cities and Society* 119: 106123. <https://doi.org/10.1016/j.scs.2025.106123>.
- Wang, H., W. Wu, L. Xiong, and F. Wang. 2024. "Is There an Environment and Economy Tradeoff for the National Key Ecological Function Area Policy in China?" *Environmental Impact Assessment Review* 104: 107347. <https://doi.org/10.1016/j.eiar.2023.107347>.
- Wei, W., Z. Shan, W. Duan, et al. 2024. "Increased Population Exposures to Extreme Precipitation in Central Asia Under 1.5°C and 2°C Global Warming Scenarios." *Geography and Sustainability* 5: 343–356. <https://doi.org/10.1016/j.geosus.2024.02.005>.
- Wen, H., J. Shang, and G. Zhang. 2025. "Does Ecological Sustainability Policy Promote Rural Income? Evidence From China's Key Ecological Function Areas." *Humanities & Social Sciences Communications* 12: 1929. <https://doi.org/10.1057/s41599-025-06207-7>.
- Wu, Z., S. Pan, G. Long, J. Jiang, X. Chang, and C. Zhang. 2020. "Connecting the Dots: Multivariate Time Series Forecasting With Graph Neural Networks. arXiv." <https://doi.org/10.48550/arXiv.2005.11650>.
- Xia, H., S. Yuan, and A. V. Prishchepov. 2023. "Spatial-Temporal Heterogeneity of Ecosystem Service Interactions and Their Social-Ecological Drivers: Implications for Spatial Planning and Management." *Resources, Conservation and Recycling* 189: 106767. <https://doi.org/10.1016/j.resconrec.2022.106767>.
- Xinhua News Agency. 2025. "What does it mean that the urbanization rate has reached 67%?" <https://www.news.cn/politics/20250508/1137a5fb4cf247adb8d4f6ea6a9b32eb/c.html>.
- Xu, K., M. Yang, W. Zhang, et al. 2025. "Assessing the Regulatory Efficacy of National Key Ecological Function Zones on Ecosystem Service Balance." *Environmental Research Letters* 20: 103001. <https://doi.org/10.1088/1748-9326/adfbfe>.
- Yang, W., J. Chen, T. Ding, X. Yan, and W. Gong. 2024. "Supply-Demand Security Assessment of Water-Energy-Food Systems: A Perspective on Intra-City Coupling and Inter-City Linkages of Ecosystem Services." *Sustainable Cities and Society* 117: 105964. <https://doi.org/10.1016/j.scs.2024.105964>.
- Yin, Z., and P. Guo. 2021. "The Impact of Targeted Poverty Alleviation Policy on Consumption: Evidence From China Household Finance Survey." *Journal of Management World* 37: 64–83. <https://doi.org/10.19744/j.cnki.11-1235/f.2021.0050>.
- Zeng, N., Z. Niu, P. Li, X. Zhu, and X. Ren. 2022. "Resistance of Grassland Productivity to Hydroclimatic Changes in the Tibetan Plateau." *Ecological Indicators* 143: 109351. <https://doi.org/10.1016/j.ecolind.2022.109351>.
- Zhang, J., C. He, Q. Huang, and L. Li. 2023. "Understanding Ecosystem Service Flows Through the Metacoupling Framework." *Ecological Indicators* 151: 110303. <https://doi.org/10.1016/j.ecolind.2023.110303>.
- Zhang, Y., H. Luo, C. Su, Y. Li, C. Xu, and H. Wang. 2024. "How Does the National Key Ecological Function Area Policy Affect the Upgrading of the Industrial Structure?—Based on the Examination of the Yellow River Basin in China." *Frontiers in Environmental Science* 12: 1446322. <https://doi.org/10.3389/fenvs.2024.1446322>.
- Zhang, Y., C. Su, and C. Xu. 2025. "How Does the National Key Ecological Function Areas Policy Affect High-Quality Economic Development?—Evidence From 243 Cities in China." *Land* 14: 345. <https://doi.org/10.3390/land14020345>.
- Zhang, Y., and Y. Wang. 2024. "How Does Urbanization Evolve Heterogeneously in Urbanized, Urbanizing, and Rural Areas of China? Insights From Ecosystem Service Value." *Geography and Sustainability* 6: 100254. <https://doi.org/10.1016/j.geosus.2024.100254>.
- Zhu, Y., C. Li, and J. Feng. 2025. "The Protected Areas System Policy Enhances Habitat Quality Conservation Effectiveness in Dongting Lake Eco-Economic Zone." *Journal of Cleaner Production* 504: 145425. <https://doi.org/10.1016/j.jclepro.2025.145425>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** Geographical division. **Appendix S2:** Ecosystem service measurement. **Appendix S3:** Spatial distribution map of county-level sample units. **Appendix S4:** Overview of multidimensional driving factors. **Appendix S5:** MTGNN hyperparameter settings. **Appendix S6:** Sensitivity analysis of HNC measurement. **Appendix S7:** MTGNN prediction and attribution sensitivity. **Figure S1:** Spatial distribution map of county-level sample units. **Figure S2:** Relationship between HNC estimates under alternative local-remote weighting schemes and the baseline HNC estimated using a 0.50/0.50 weighting scheme. The black line represents the 1:1 reference line; the red line represents the fitted linear relationship. **Figure S3:**

Distribution of HNC deviations under alternative local-remote weighting schemes relative to the 0.50/0.50 baseline specification. **Figure S4:** Stability of MTGNN feature rankings under alternative HNC weighting schemes. (a) Trajectories of temporally aggregated feature ranks. (b) Heat map of ranking stability for base variables. **Figure S5:** Stability of IG value structure and temporal-lag contributions under alternative HNC weighting schemes. (a) Correlations of temporally aggregated IG vectors. (b) Relative contributions from the current and lagged time steps. **Table S1:** Regions of China and provinces included. **Table S2:** HNC statistics under alternative local-remote weights. **Table S3:** MTGNN test performance across HNC weighting schemes.